

EXTREME NETWORKS

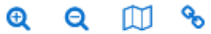
XIQ-SE workflow: Fabric MultiArea Migrate

Ludovico Stevens, December 2022

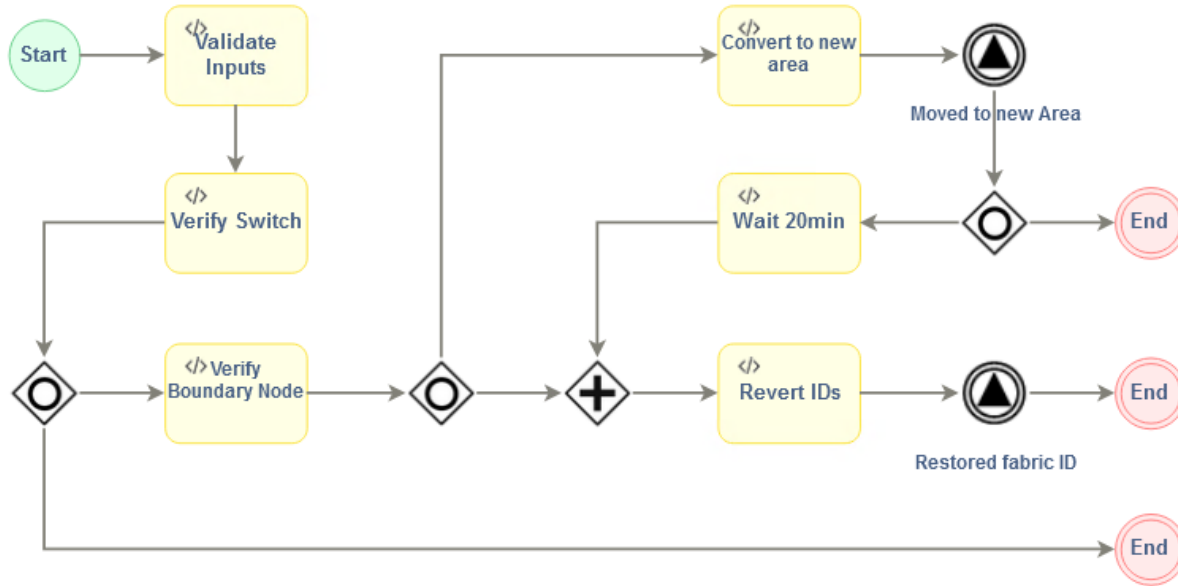
The workflow



/Workflows/Ludovico/Fabric MultiArea Migrate



Verify Save Run



Workflow allows changing the ISIS area of a switch in a multi-area design where nodes need to be migrated to new areas

Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:

49.0000

Target ISIS Area:

49.0001

Mgmt IP of Area Boundary Nodes:

The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:

20.0.20.76

Fabric ID mask manipulation Notes:

Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:

replace

System-id mask:

0000.F000.0000

System-id mask replace value:

2

System-id in new area:

Keep modified ID

Nick-name mask manipulation:

XOR

Nick-name mask:

8.00.00

Nick-name mask replace value:

Nick-name in new area:

Revert to original ID after 20min

ISIS Hello Authentication Notes:

If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:

enable

ISIS Hello Authentication key:

....

Running the workflow



- **Run the workflow against a single switch, if the switch is standalone and not part of a vIST cluster**

The screenshot displays the 'Devices' tab in the management console. A table lists various network devices, including VSP4450-1, VSP4450-2, VSP4900-3, VSP4900-4, X435-8P-2T-W, X435-8P-2T-W, X440G2-12p-10G4, and X460G2-2. The 'VSP4900-3' device is highlighted, and a context menu is open over it. The menu includes options like 'Device View', 'Terminal', 'WebView', 'FlexView', 'More Views', 'Configure...', 'Compass Search...', 'Rediscover', 'Clear Alarms...', 'Upgrade Firmware...', 'Add to Device Group...', 'More Actions', 'Archives', 'Tasks', 'Maps', 'Network', 'Policy', and 'CLI Commands...'. The 'Fabric MultiArea Migrate' option is highlighted in red. To the right, a table shows device details for VSP4900-48P, including 'Device Type', 'Family', and 'Firmware'.

Status	Name	Site	IP Address	Poll Status
▲	VSP4450-1	/World/CTC/Readin...	20.0.20.41	Available: 1..
▲	VSP4450-2		20.0.20.42	Available: 1..
▲	VSP4900-3		20.0.20.43	Available: 1..
▶	VSP4900-4		20.0.20.44	Available: 1..
▲	X435-8P-2T-W		20.0.205.11	Available: 1..
▲	X435-8P-2T-W		20.0.206.11	Available: 1..
●	X440G2-12p-10G4		20.0.204.11	Available: 1..
●	X460G2-2		20.0.204.12	Available: 1..

Device Type	Family	Firmware
4450GSX-PWR+	VSP Series	8.5.0.0
4450GSX-PWR+	VSP Series	8.5.0.0
4900-48P	VSP Series	8.9.0.0_B011
4900-48P	VSP Series	8.8.0.0
8P-2T-W	Summit Se...	31.3.1.3-pa...
8P-2T-W	Summit Se...	31.3.1.3-pa...
G2-12p-10G4	Summit Se...	31.6.1.3
G2-24t-10G4	Summit Se...	31.5.1.6

Running the workflow



- **Run the workflow against two switches, if the switches form a vIST cluster**
- **The workflow will error if launched against a single vIST peer switch, or if the two selected switches are not part of the same vIST cluster, or if more than 2 switches are selected**

The screenshot shows the Cisco DNA Center interface with the 'Devices' tab selected. The table lists two switches, VSP8200-1 and VSP8200-2, both with status 'Available'. A context menu is open over the selected switches, showing various actions. The 'Fabric MultiArea Migrate' option is highlighted with a red box.

Status	Name	Site	IP Address	Poll Status	Poll Details	Device Type
▶	VSP8200-1	/World/CTC/Readin...	20.0.20.31	Available	...	VSP-8284XSQ
▶	VSP8200-2	/World/CTC/Readin...	20.0.20.32	Available	...	VSP-8284XSQ

Context Menu Options:

- FlexView
- More Views
- Configure...
- Compass Search...
- Rediscover
- Clear Alarms...
- Upgrade Firmware...
- Add to Device Group...
- More Actions
- Archives
- Tasks
- Maps
- Network
- Policy
- Access Control
- Config
- Example
- Macro
- Provisioning
- System
- VLAN
- CLI Commands...
- Apply Config Template
- CLI Custom Action - XOS SSL
- Delete Insight VMs
- Deploy Insight VM
- Disable IQAgent
- Enforce Config after Onboard
- Fabric Attach Enforce
- Fabric Connect Enforce
- Fabric MultiArea Migrate
- Move to CLIP Mgmt IP
- Onboard Mgmt CLIP
- Onboard Mgmt VLAN
- Onboard VSP
- SMLT Pair Enforce
- by Marlon - Onboard VSP
- deliver VOSS license
- onboard CLI
- onboard EXOS

Running the workflow

- **Provide the workflow inputs**
- Inception & target area ids
- Mgmt IP of 1 Multi-Area boundary node
 - This is used to check modified fabric IDs in both inception and target areas
- Select the type of ID manipulation to use for system-id and nickname
 - XOR: Mask determines which bits in the ID will be flipped
 - Replace: Mask determines which bits in the ID will be replaced with value provided
 - If a vIST cluster, the system-id manipulation will also get applied to SMLT Virt BMAC and SMLT Peer System-id
- Specify if either the Nickname or System-id are to be reverted to original ID after 20 minutes in new target area
 - If a vIST cluster, the system-id reversion will also get applied to SMLT Virt BMAC and SMLT Peer System-id
- Optionally, the ISIS hello authentication can be changed during the area move; if this is desired enable the option and provide a auth key
 - This will apply to all NNI interfaces, including MLT and FE ones
 - If the option is enabled, and no key is provided, this will result in deleting ISIS Hello auth from the interfaces

Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:

49.0000

Target ISIS Area:

49.0001

Mgmt IP of Area Boundary Notes:

The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:

20.0.20.76

Fabric ID mask manipulation Notes:

Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:

replace

System-id mask:

0000.F000.0000

System-id mask replace value:

2

System-id in new area:

Keep modified ID

Nick-name mask manipulation:

XOR

Nick-name mask:

8.00.00

Nick-name mask replace value:

Nick-name in new area:

Revert to original ID after 20min

ISIS Hello Authentication Notes:

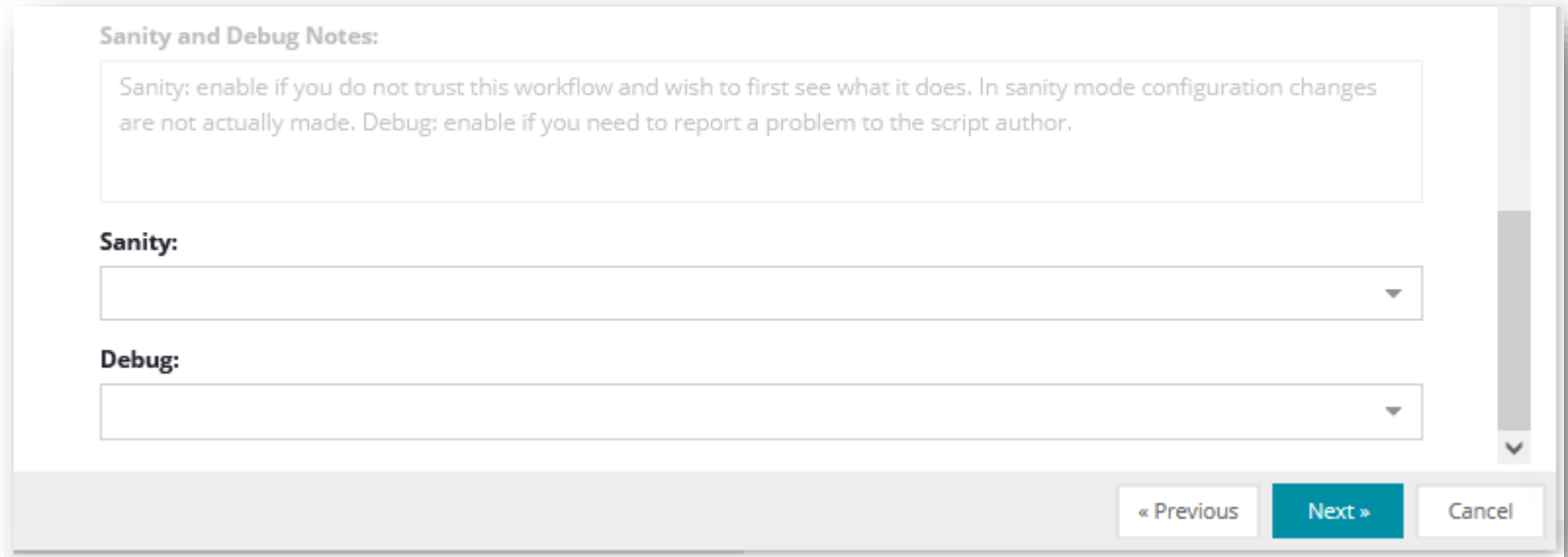
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:

enable

ISIS Hello Authentication key:





The dialog box titled "Sanity and Debug Notes" contains a text area with instructions: "Sanity: enable if you do not trust this workflow and wish to first see what it does. In sanity mode configuration changes are not actually made. Debug: enable if you need to report a problem to the script author." Below this are two dropdown menus labeled "Sanity:" and "Debug:". At the bottom right are three buttons: "« Previous" (disabled), "Next »" (active), and "Cancel".

- **Sanity and Debug inputs**

- Sanity: Enable this initially, when testing the workflow. The workflow will run but will not make any config changes on the switch nor any API writes to XIQ-SE. Allows to understand what the workflow does and how it works, without actually doing the changes.
- Debug: If the workflow is failing, enable this, so get extra debug in the activity logs

Running the workflow



- **Hit Run**

A screenshot of a software dialog box titled "Run Workflow - Fabric MultiArea Migrate". The dialog has a standard Windows-style title bar with maximize, minimize, and close buttons. Below the title bar, the text "Schedule Task" is displayed. Underneath, there is a section titled "Task Details". This section contains three input fields: "Task Name:" with the value "Scheduled Task - 12/8/2022 10:37:56 AM", "Description:" which is empty, and "Enabled:" with an unchecked checkbox. At the bottom of the dialog, there is a row of four buttons: "« Previous", "Run" (highlighted in blue), "Save Task" (with a floppy disk icon), and "Cancel".

Run Workflow - Fabric MultiArea Migrate

Schedule Task

Task Details

Task Name: Scheduled Task - 12/8/2022 10:37:56 AM

Description:

Enabled: ☐

« Previous Run Save Task Cancel

Wait for workflow completion



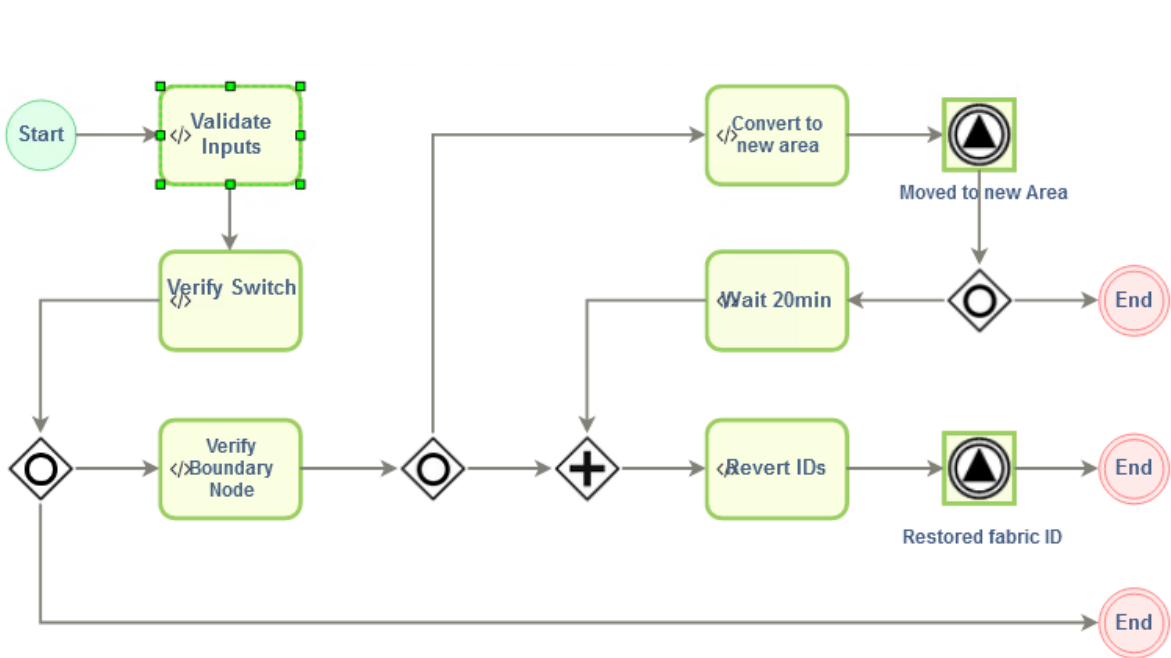
Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message	Path
✓	12/8/2022 9:18:17 ...	Fabric MultiArea Mi...	52	Workflow Designer ...	2	Istevens	12/8/2022 9:18:45 ...	Configured RADIUS and EAPoL on devices 20.0.20.31, 20.0.20.32	/Workflows/Ludovico/Fabric MultiArea Migrate

Graph View Table View



Stop Workflow Show Output Show Variables



Devices Grid

Show Output

Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	20.0.20.31		12/8/2022 9:1...	12/8/2022 9:1...	Validated ...
SUCCESS	20.0.20.32		12/8/2022 9:1...	12/8/2022 9:1...	Validated ...

Inspect Workflow execution



Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message	Path
✓	12/8/2022 9:18:17 ...	Fabric MultiArea Mi...	52	Workflow Designer ...	2	Istevens	12/8/2022 9:18:45 ...	Configured RADIUS and EAPoL on devices 20.0.20.31, 20.0.20.32	/Workflows/Ludovico/Fabric MultiArea Migrate

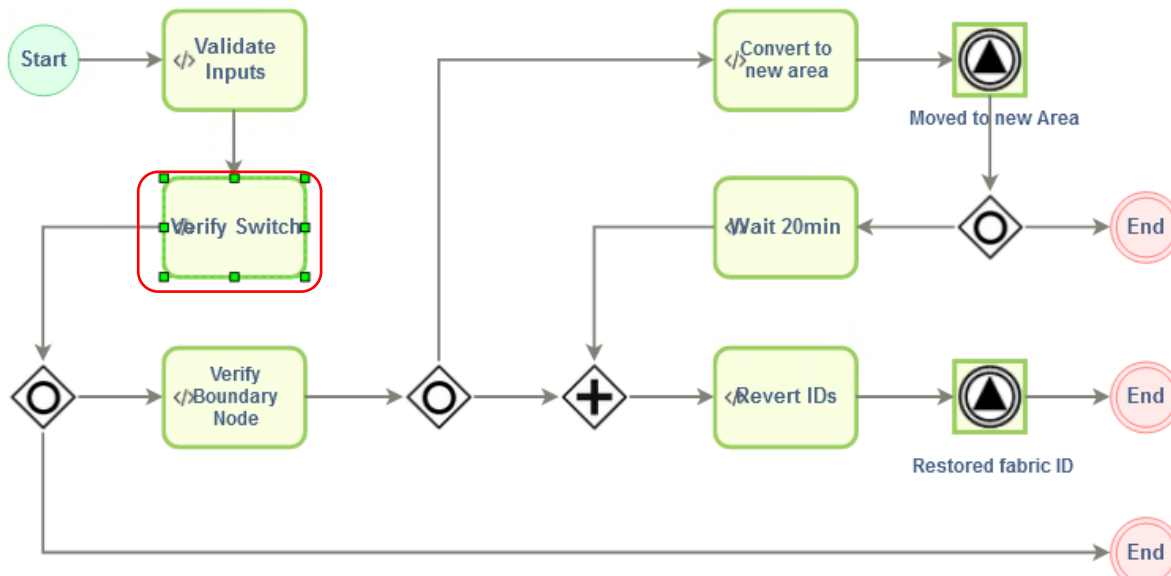
Graph View Table View



Stop Workflow Show Output Show Variables

Devices Grid

Show Output



Output - 20.0.20.31

```
=====
=====
ISIS Area Address
=====
=====
ADDRESS                ORIGIN  AREA    AREA-NAME
-----
49.0000                config HOME
VSP8200-1:1#
Current and modified fabric IDs to use in target area 49.0002 for switch 20.0.20.31:
- Current System-id = 82bb.0000.3100
- Modified System-id = 82bb.2000.3100
- Current Nickname = 0.00.31
- Modified Nickname = 8.00.31
- Current SMLT Virt BMAC = 82:bb:00:00:31:ff
- Modified SMLT Virt BMAC = 82:bb:20:00:31:ff
- Current SMLT Peer System-id = 82bb.0000.3200
- Modified SMLT Peer System-id = 82bb.2000.3200
SANITY - NBI Mutation:
```

Select activity, then “show Output”
Activity which fail, will be in red

Workflow Events recorded



Alarms Alarm Configuration Events Event Configuration										
All Type: Console View Export to CSV										
Severity	Event Type	Category	Date/Time	Source	Subcomponent	Client	User	Type	Event	Information
	Console	Workflows	12/8/2022 10:11:17 AM	[20.0.20.43]	Workflow: Fabric MultiArea Migrate	---	Istevens	Event	Fabric IDs reverted on device	Reverted fabric IDs to original values for device(s) [20.0.20.43]
	Console	Workflows	12/8/2022 10:11:12 AM	[20.0.20.43]	Workflow: Fabric MultiArea Migrate	---	Istevens	Event	Device moved to new ISIS area	Device(s) [20.0.20.43] moved from ISIS area 49.0000 to 49.0002

EXTREME NETWORKS

How to migrate to SPB Multi-Area from a single Area Fabric using XIQ-SE workflow

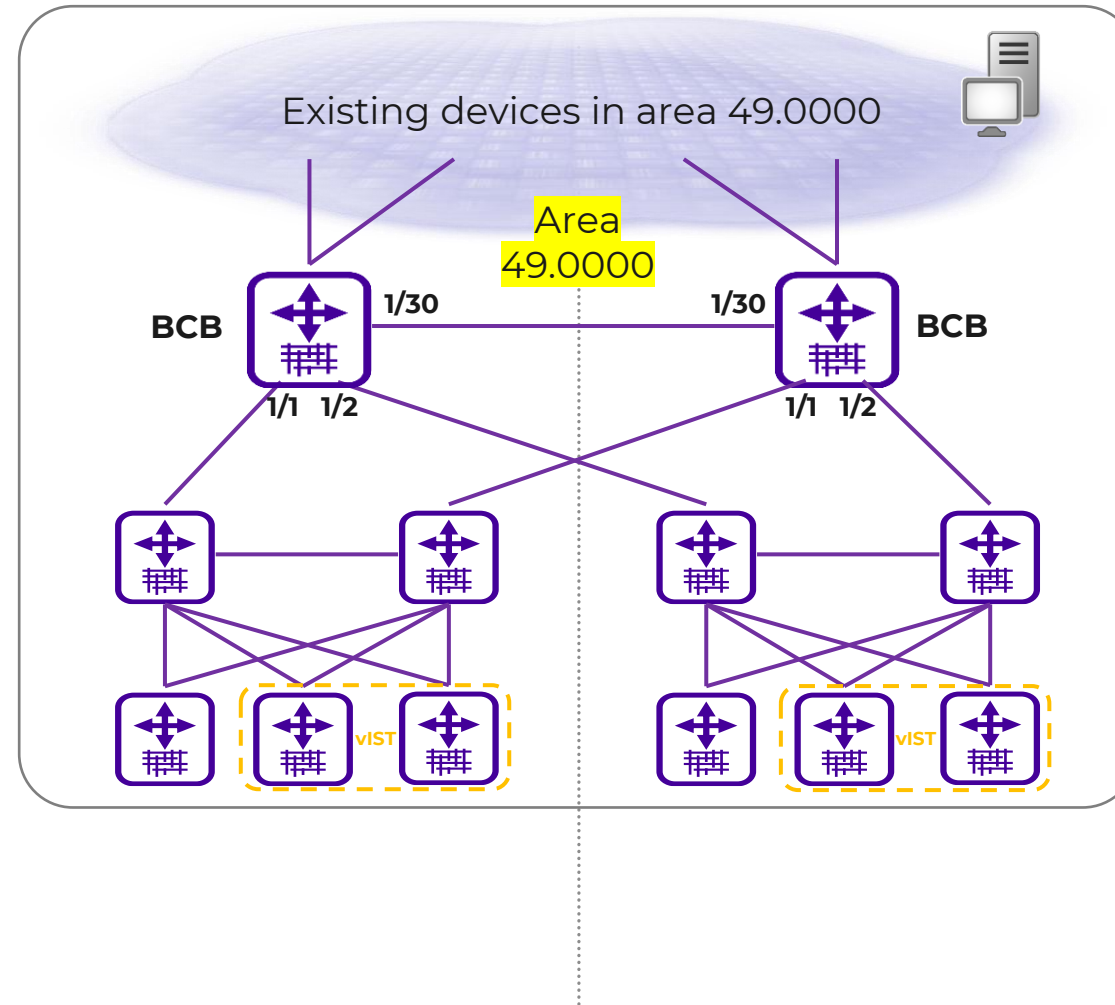
Ludovico Stevens, Technical Marketing Engineering

December 2022

Inception



- All fabric nodes and all NNI links belong to one and only ISIS area 49.0000

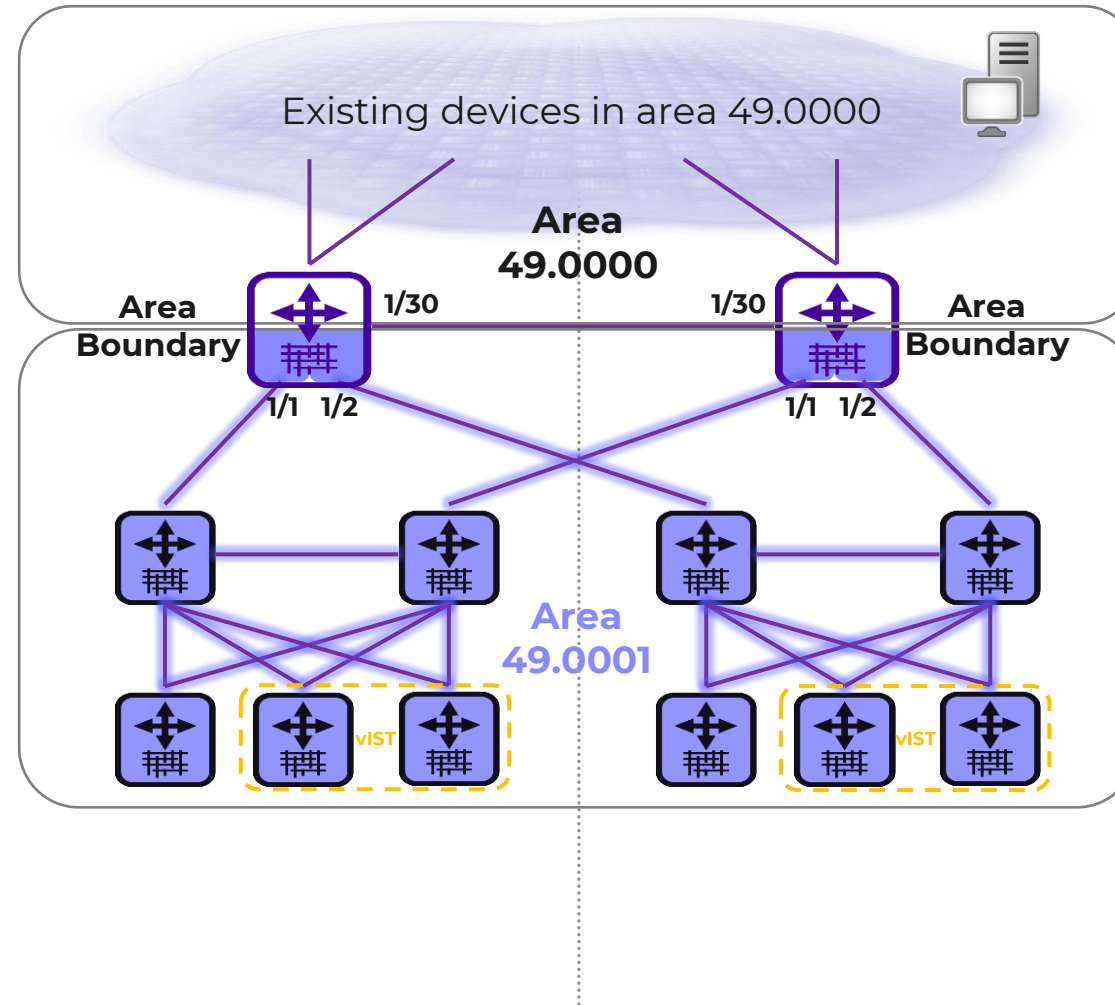


- Management station located on existing area
- All configuration performed from management station (using SSH)

Target



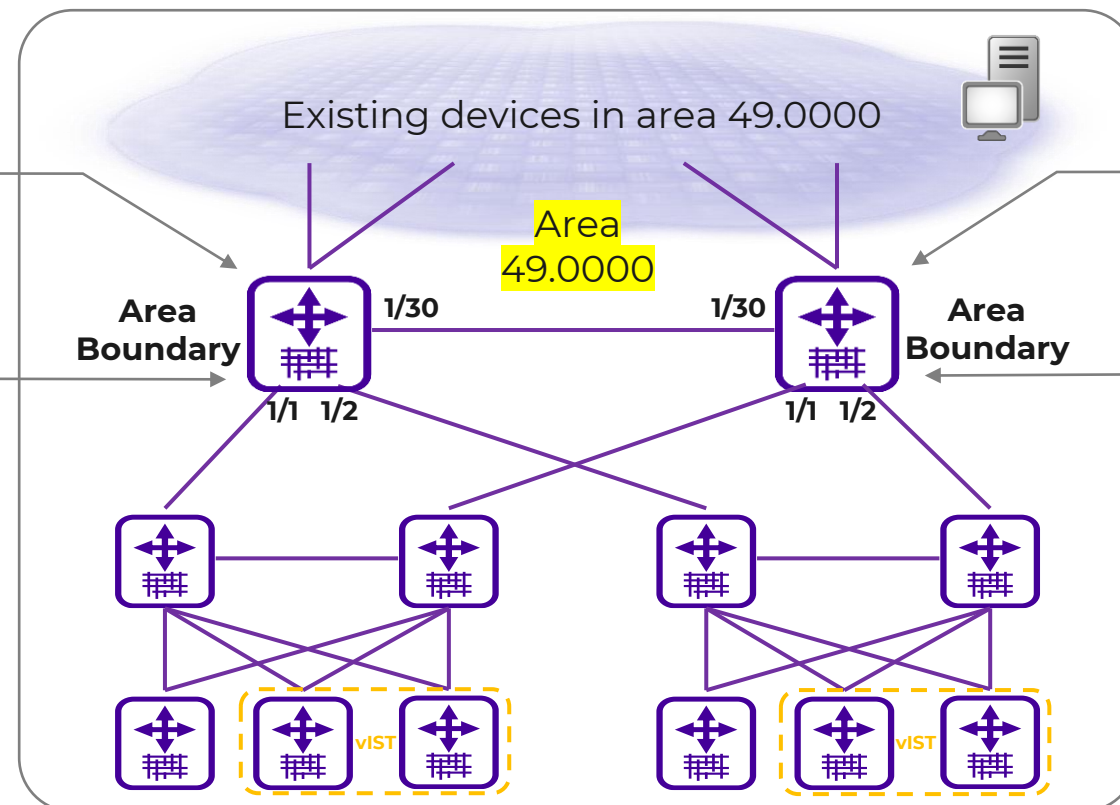
- Bottom fabric nodes are moved to a new access area 49.0001
- The blue glowing links are all operating in the new area 49.0001



- All migration steps performed from management station
- Goal is that local access or serial port access should NOT be required on the nodes being moved to the new area


```
router isis
  spbm 1
  spbm 1 nick-name 1.00.71
  spbm 1 b-vid 4051-4052 primary 4051
  system-id 02bb.0000.7100
  manual-area 49.0001
exit
vlan create 4051 name "B-VLAN-1" type spbm-bvlan
vlan create 4052 name "B-VLAN-2" type spbm-bvlan
router isis enable
```

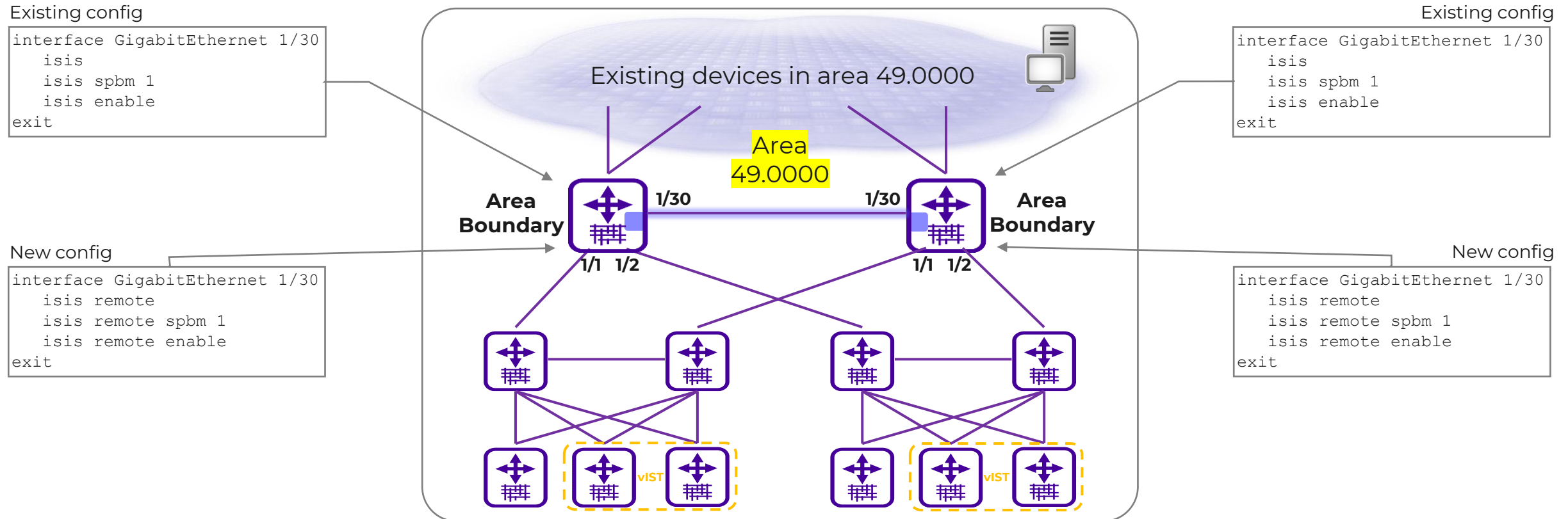
```
router isis
  area-name core-area
  area-vnode sys-name Vnode-49.0000
exit
router isis remote
  spbm 1 nick-name 2.00.71
  system-id e2bb.0000.7100
  manual-area 49.0001
  area-name access-area
  area-vnode sys-name Vnode-49.0001
exit
router isis remote enable
```



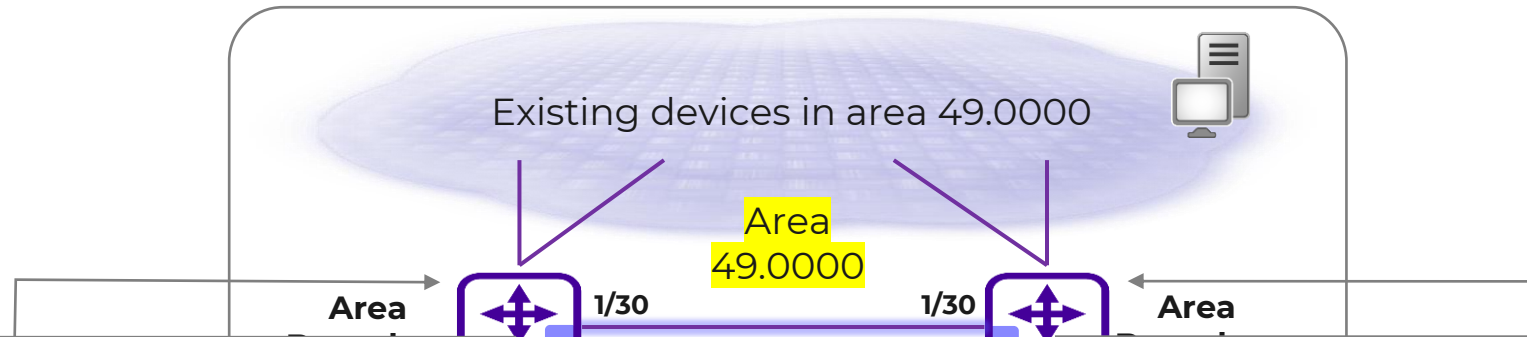
```
router isis
  spbm 1
  spbm 1 nick-name 1.00.72
  spbm 1 b-vid 4051-4052 primary 4051
  system-id 02bb.0000.7200
  manual-area 49.0001
exit
vlan create 4051 name "B-VLAN-1" type spbm-bvlan
vlan create 4052 name "B-VLAN-2" type spbm-bvlan
router isis enable
```

```
router isis
    area-name core-area
    area-vnode sys-name Vnode-49.0000
exit
router isis remote
    spbm 1 nick-name 2.00.72
    system-id e2bb.0000.7200
    manual-area 49.0001
    area-name access-area
    area-vnode sys-name Vnode-49.0001
exit
router isis remote enable
```

- It is assumed that both boundary nodes were already configured for and members of area 49.0000
- We therefore configure the new access area as the remote area 49.0001
- It is worth naming the existing and new area appropriately as well as the Virtual-Node sys-name



- If the area boundary nodes share an NNI link, it is important to have both a home area ISIS adjacency and a remote area adjacency on that link
- If the area boundary nodes do not share an NNI link, there are alternative multi-area designs
 - A multi-area virtual link would have to be created, over the home area, allowing the boundary nodes to see each other in the new remote area
 - Multi-area virtual links require VOSS 9.0.3.0 and VSP7400 or 7720 or 7520 boundary nodes



VSP7432-1:1(config)#% show isis

ISIS General Info

```

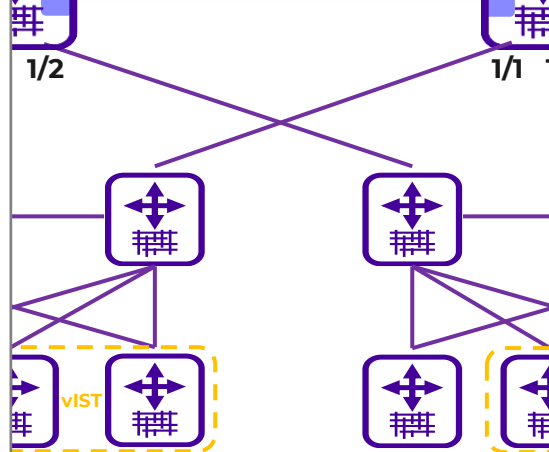
AdminState : enabled
RouterType : Level 1
System ID : 02bb.0000.7100
Max LSP Gen Interval : 900
Metric : wide
Overload-on-startup : 20
Overload : false
Csnp Interval : 10
PSNP Interval : 2
Rxmt LSP Interval : 5
spf-delay : 100
Router Name : VSP7432-1
ip source-address :
ipv6 source-address :
ip tunnel source-address :
Tunnel vrf :
ip tunnel mtu :
Num of Interfaces : 3
Num of Area Addresses : 1
Inband Mgmt Clip Ip :
backbone : disabled
Dynamically Learned Area :
FAN Member : No
Multi-Area OperState : designated
    
```

VSP7432-1:1(config)#[ma]% show isis remote

ISIS Remote Area Info

```

AdminState : enabled
System ID : e2bb.0000.7100
Num of Interfaces : 3
Num of Area Addresses : 1
Dynamically Learned Area :
Multi-Area OperState : designated
    
```



- Verify that boundary nodes are in OperState Designated and Non-designated

VSP7432-2:1(config)#% show isis

ISIS General Info

```

AdminState : enabled
RouterType : Level 1
System ID : 02bb.0000.7200
Max LSP Gen Interval : 900
Metric : wide
Overload-on-startup : 20
Overload : false
Csnp Interval : 10
PSNP Interval : 2
Rxmt LSP Interval : 5
spf-delay : 100
Router Name : VSP7432-2
ip source-address :
ipv6 source-address :
ip tunnel source-address :
Tunnel vrf :
ip tunnel mtu :
Num of Interfaces : 3
Num of Area Addresses : 1
Inband Mgmt Clip Ip :
backbone : disabled
Dynamically Learned Area :
FAN Member : No
Multi-Area OperState : non-designated
    
```

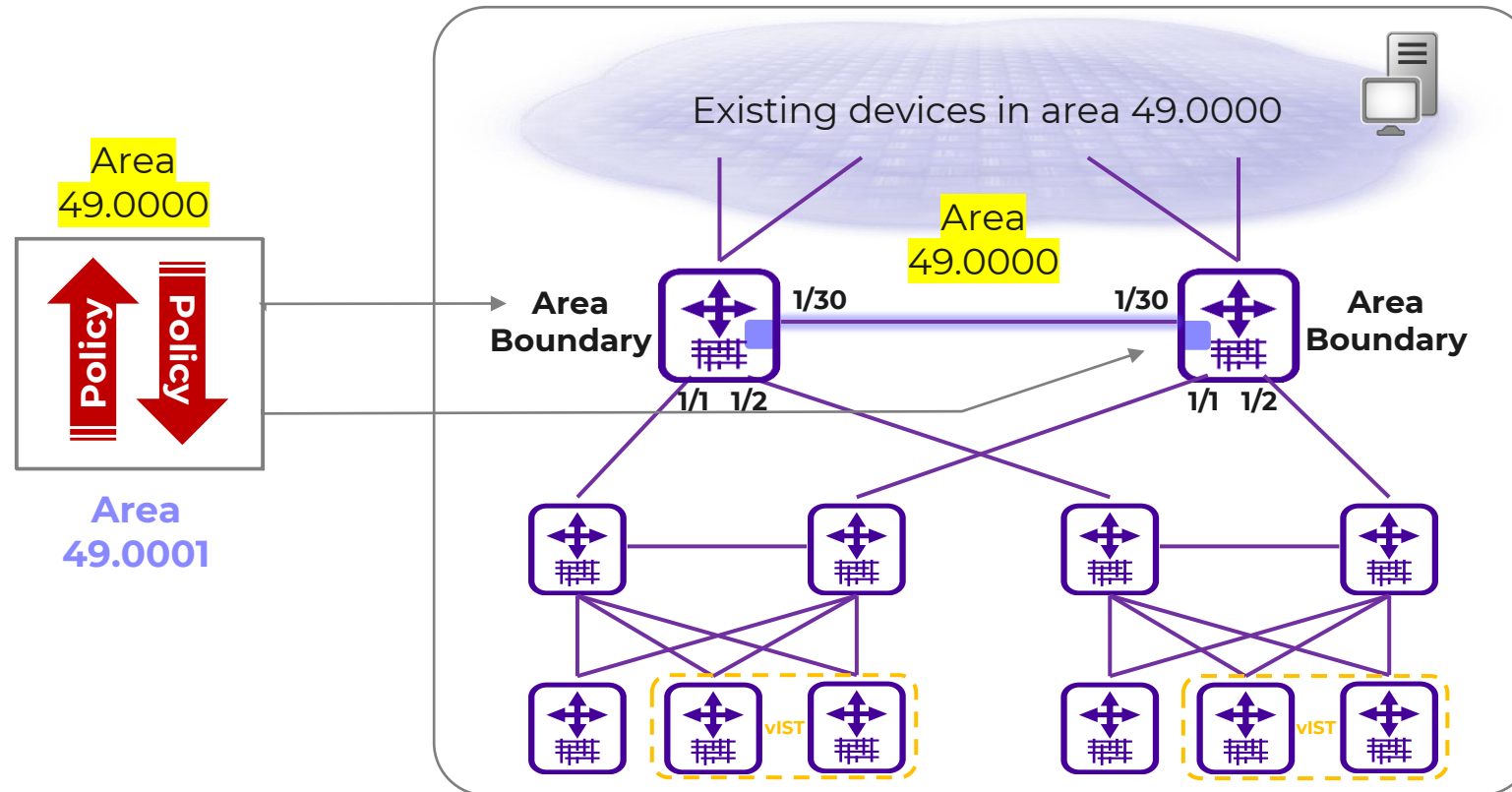
VSP7432-2:1(config)#% show isis remote

ISIS Remote Area Info

```

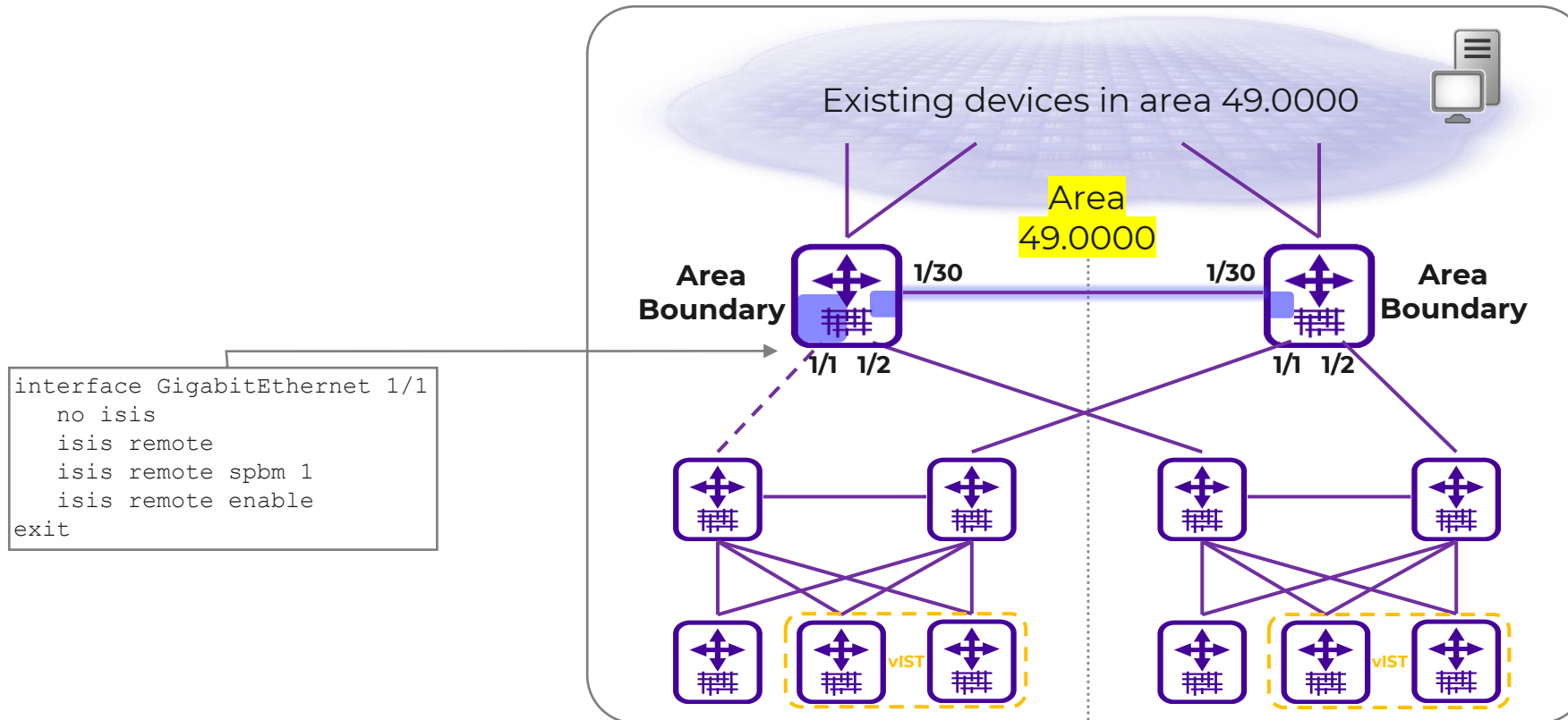
AdminState : enabled
System ID : e2bb.0000.7200
Num of Interfaces : 3
Num of Area Addresses : 1
Dynamically Learned Area :
Multi-Area OperState : non-designated
    
```

Multi-area service policies



- Apply the multi-area redistribution of required services between the existing area 49.0000 and the new access area 49.0001
- This can apply to IP Shortcuts, L2VSNs, L3VSNs, Multicast (both dynamic & static), DVR Backbone and Fabric RSPAN
- Config omitted in this deck

Prepare first boundary remote Area NNI



- Delete home area ISIS and re-create as remote area ISIS NNI
 - Note that it is not possible to have both ISIS Home and Remote configured on these same NNI interfaces (this can only work on direct NNI links between the two boundary nodes)

- This will cause the ISIS adjacency on the link to go down and not come back up, because the access distribution node at the other end is still configured for area 49.0000, not area 49.0001
- An area mis-match message will appear in the log file of both nodes sharing that link
- Access nodes can still be reached on area 49.0000 via other boundary node's NNI

Run Fabric MultiArea Migrate workflow on 1st node to move to new area



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Notes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
20.0.20.76

Fabric ID mask manipulation Notes:
Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

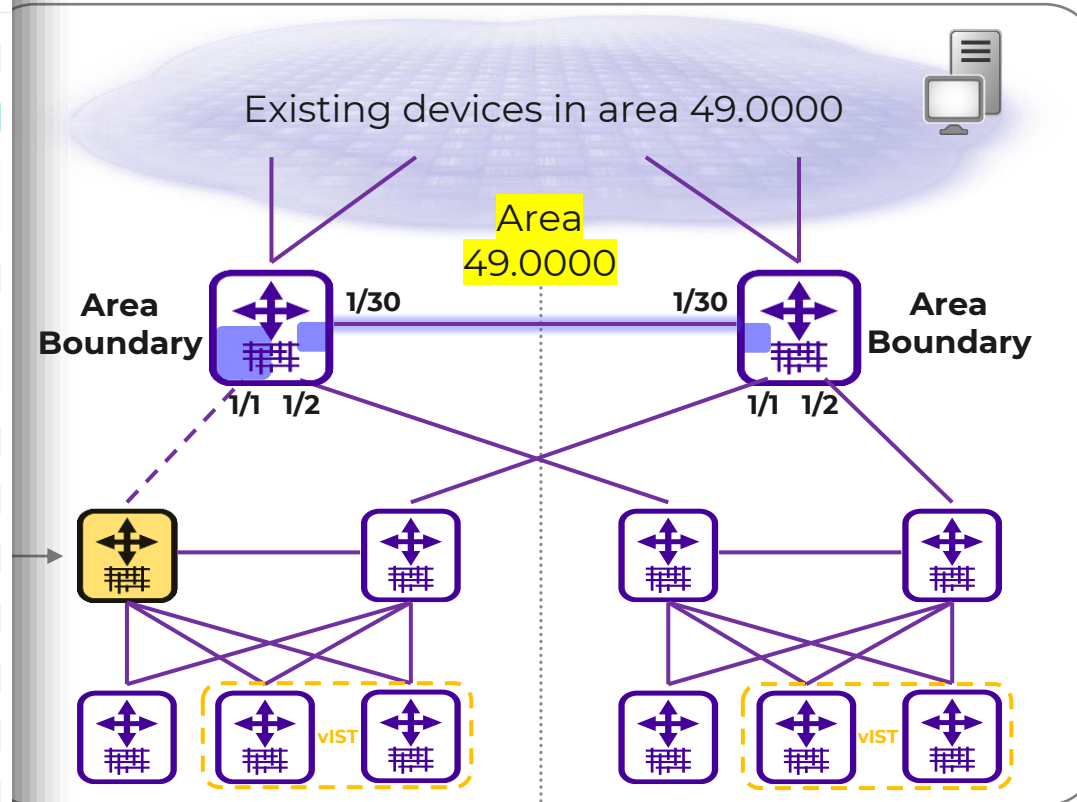
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:



- Run the workflow against the switch
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
- Run the workflow
 - Changes are applied dynamically without a reboot

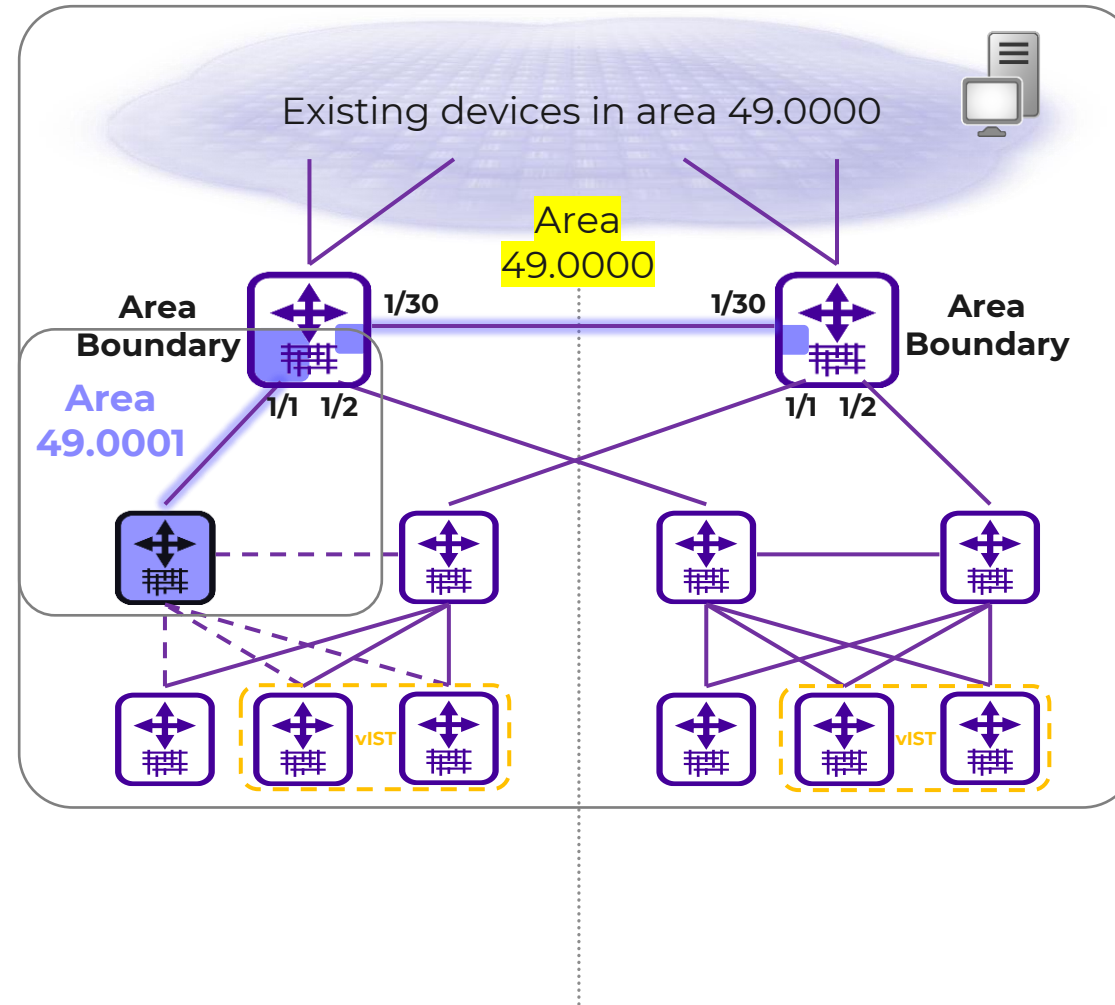
Important to realize that the node's existing nick-name and system-id are already advertised in the home area LSDB and will remain there for up to 20 minutes after the node is removed from the area 49.0000

The node must therefore be given a new nick-name and system-id to operate in the new area

1st node comes online in new access area



- When the node comes back online, it will have an adjacency in the new area 49.0001 to the boundary node
- But all other ISIS adjacency it had will be down, as its old neighbours are still in area 49.0000



- If it was desired to revert the node back to its original nickname and system-id the workflow will wait 20 minutes and then revert to the original IDs
- Should the workflow be unable to re-connect to revert the fabric IDs, it will be sufficient to manually re-run the same workflow (with same inputs) a second time and it will resume from where it left

Workflow run against single switch



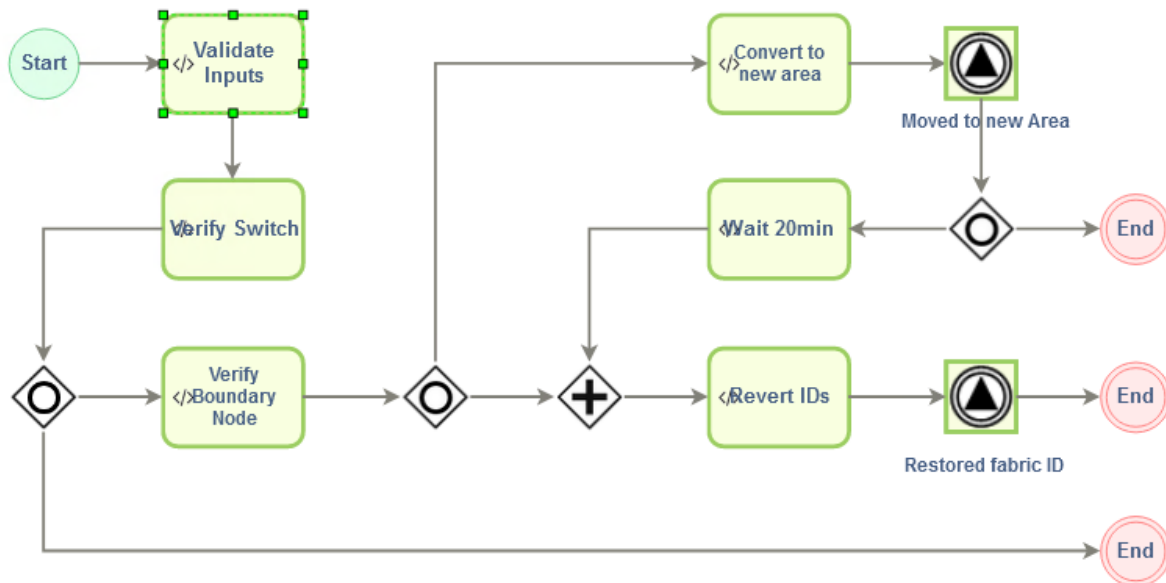
Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message	Path
✓	12/8/2022 10:10:51...	Fabric MultiArea Mi...	58	Workflow Designer ...	1	Istevens	12/8/2022 10:11:17...	Configured RADIUS and EAPoL on device 20.0.20.43	/Workflows/Ludovico/Fabric MultiArea Migrate

Graph View

Table View

Stop Workflow Show Output Show Variables



Devices Grid

Show Output

Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	20.0.20.43		12/8/2022 10:...	12/8/2022 10:...	Validated ...

Run Fabric MultiArea Migrate workflow on 2nd node to move to new area



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Notes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
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System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

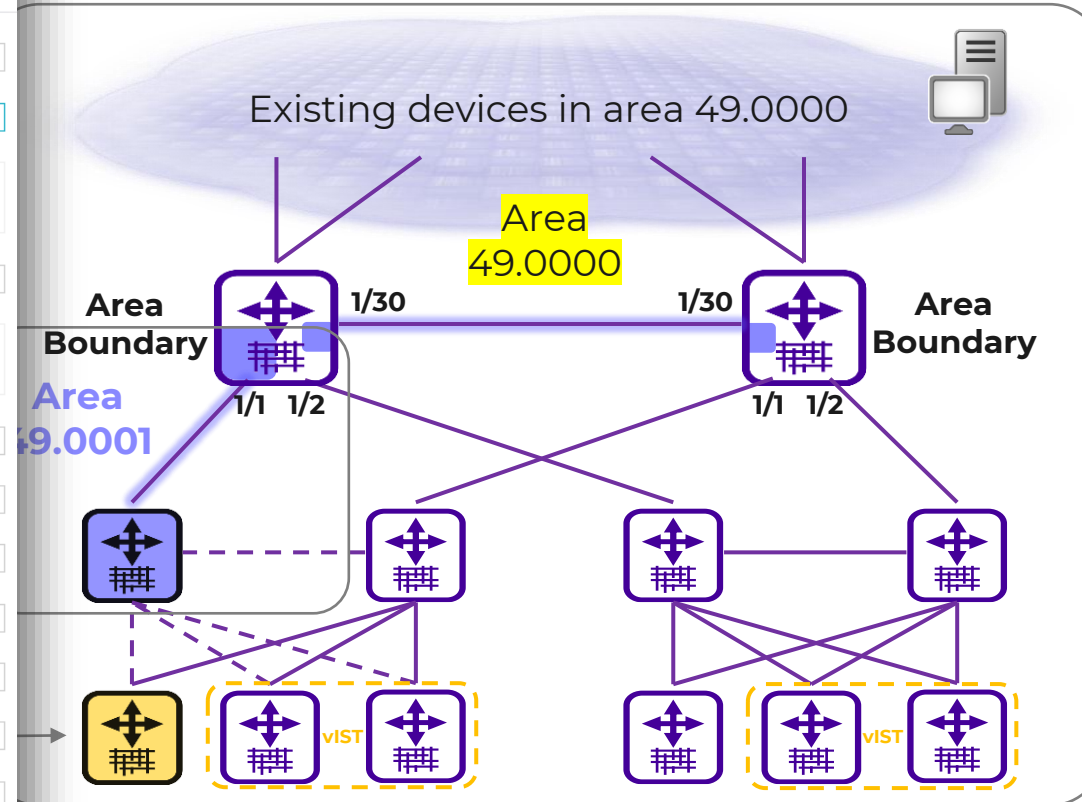
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:

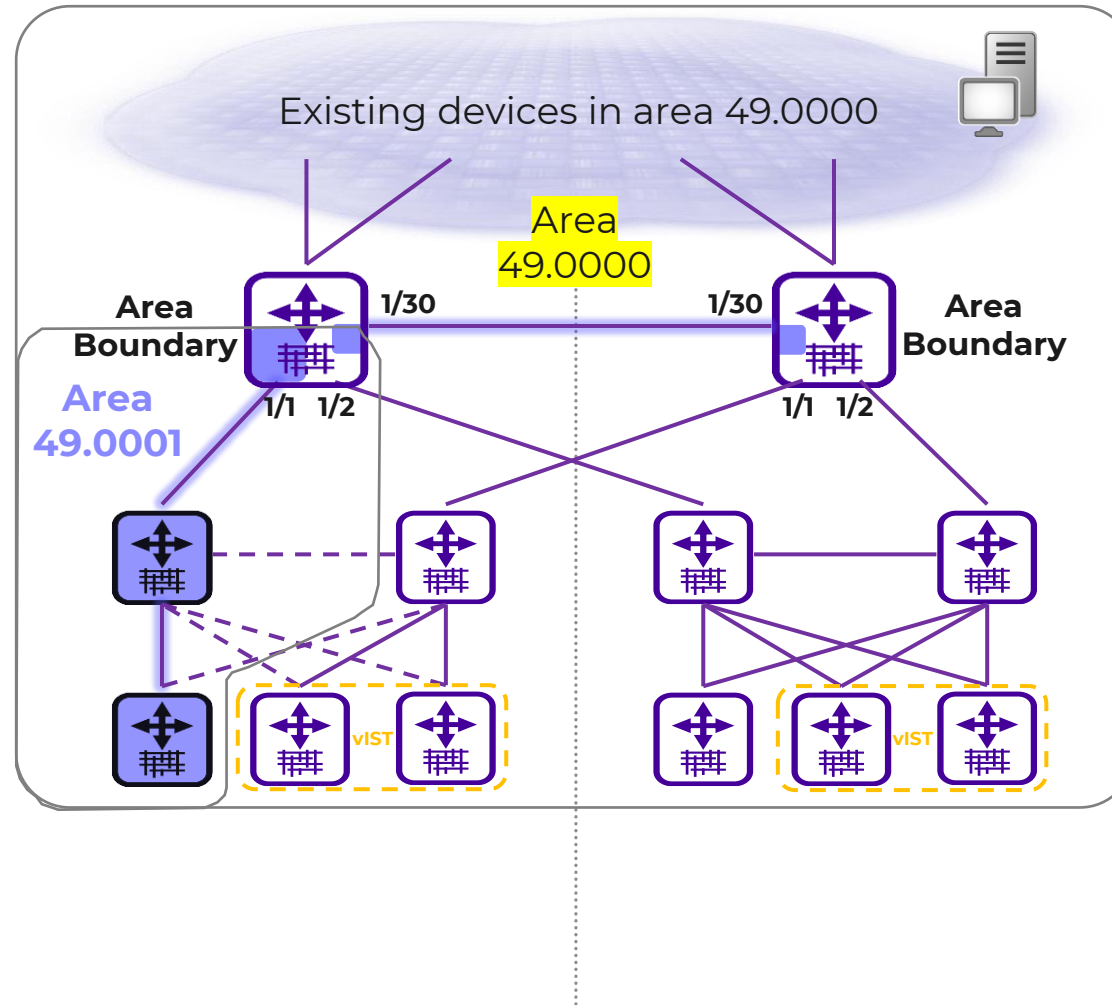


- Run the workflow against the switch
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
- Run the workflow
 - Changes are applied dynamically without a reboot

2nd node comes online in new access area



- When the node comes back online, it will have an adjacency in the new area 49.0001 going into the 1st boundary node
- But all other ISIS adjacency it had will be down, as its old neighbours are still in area 49.0000



- If it was desired to revert the node back to its original nickname and system-id the workflow will wait 20 minutes and then revert to the original IDs
- Should the workflow be unable to re-connect to revert the fabric IDs, it will be sufficient to manually re-run the same workflow (with same inputs) a second time and it will resume from where it left

Run Fabric MultiArea Migrate workflow on vIST Cluster nodes



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Nodes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
20.0.20.76

Fabric ID mask manipulation Notes:
Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

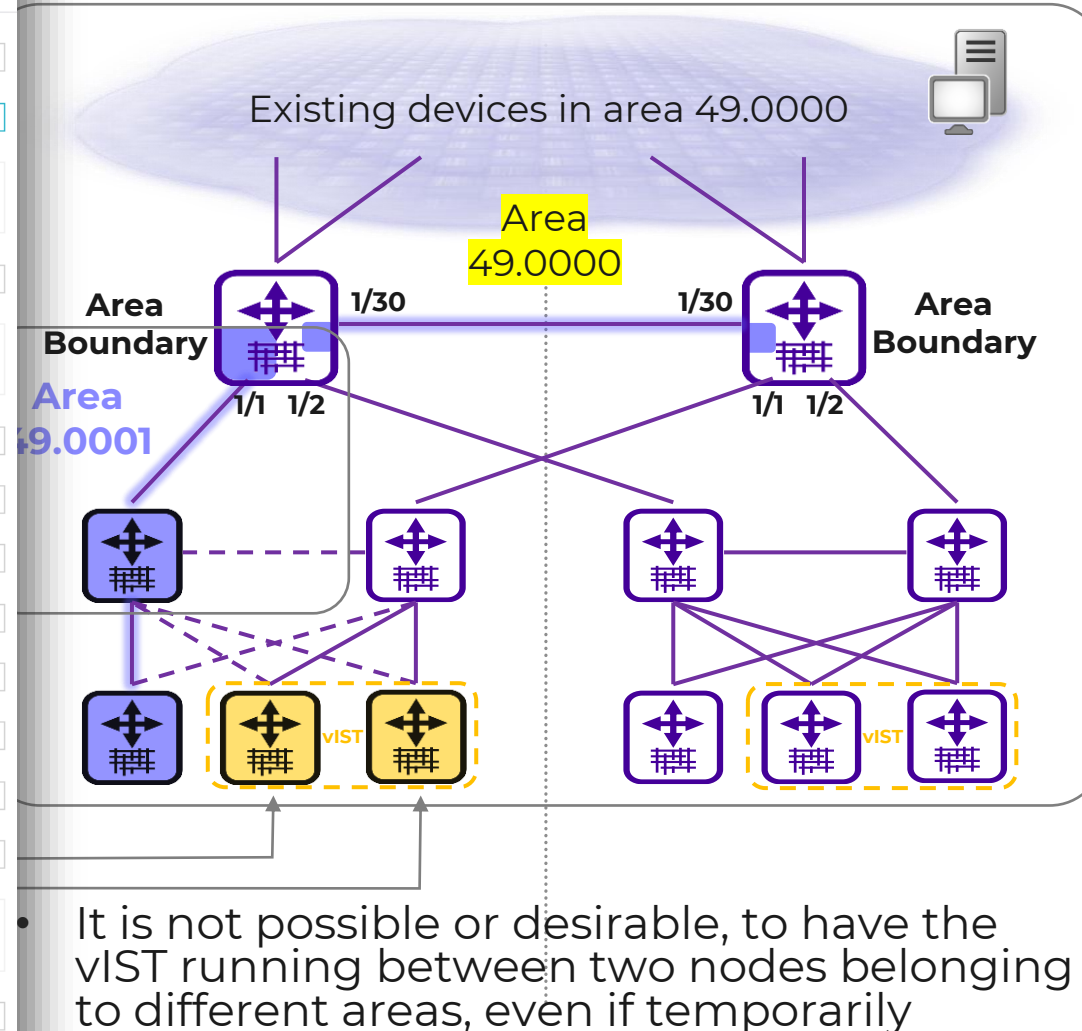
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:

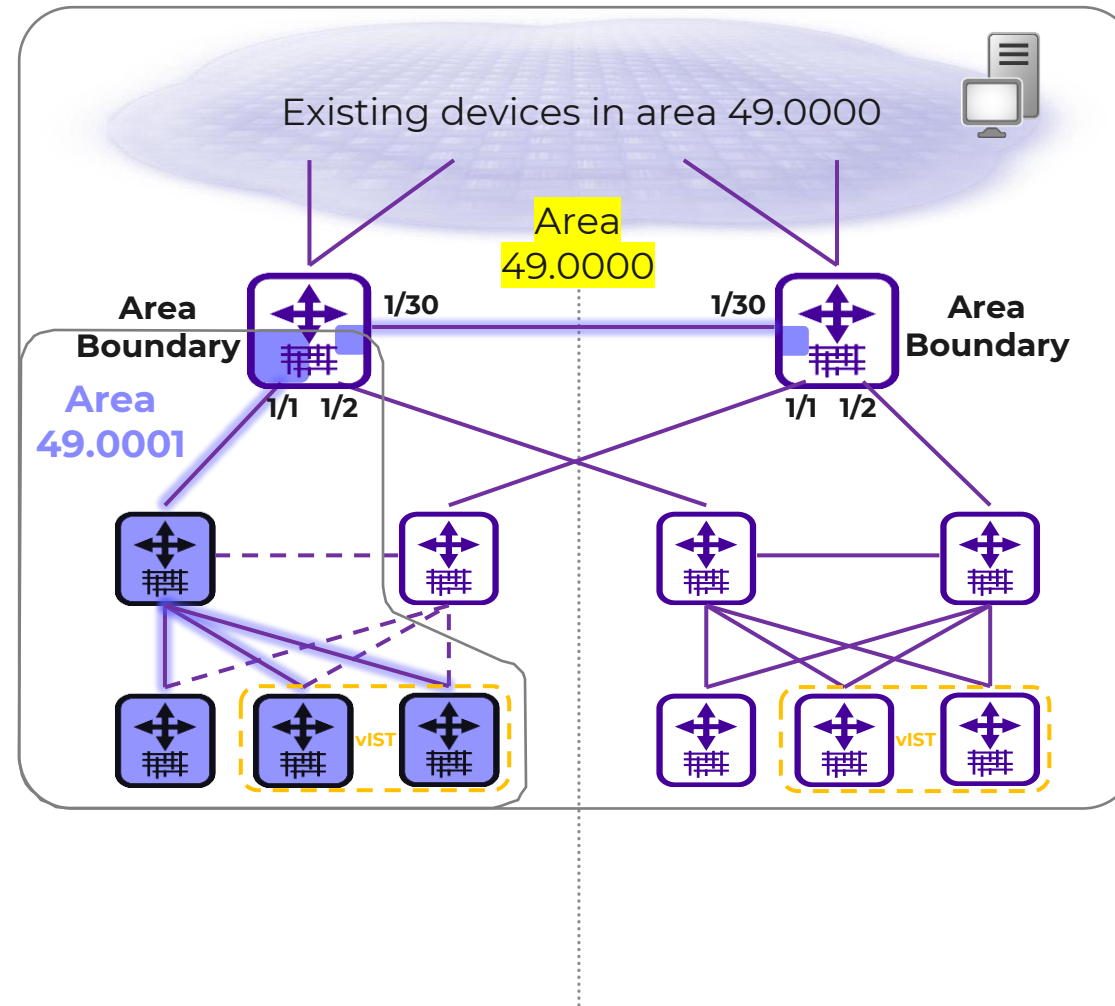


- Run the workflow against both vIST peer switches
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
 - SMLT-Peer-Sys-id to match above
 - SMLT-Virt-BMAC
 - Manual-area
- Run the workflow
 - Changes are applied dynamically without a reboot

vIST cluster nodes come online in new access area



- When the vIST cluster comes back online, it will have adjacencies in the new area 49.0001 going into the 1st boundary node
- But all other ISIS adjacency it had will be down, as its old neighbours are still in area 49.0000



- If it was desired to revert the vIST nodes back to their original nick-name and system-id the workflow will wait 20 minutes and then revert to the original IDs
- Should the workflow be unable to re-connect to revert the fabric IDs, it will be sufficient to manually re-run the same workflow (with same inputs) a second time and it will resume from where it left

Workflow run against vIST pair of switches



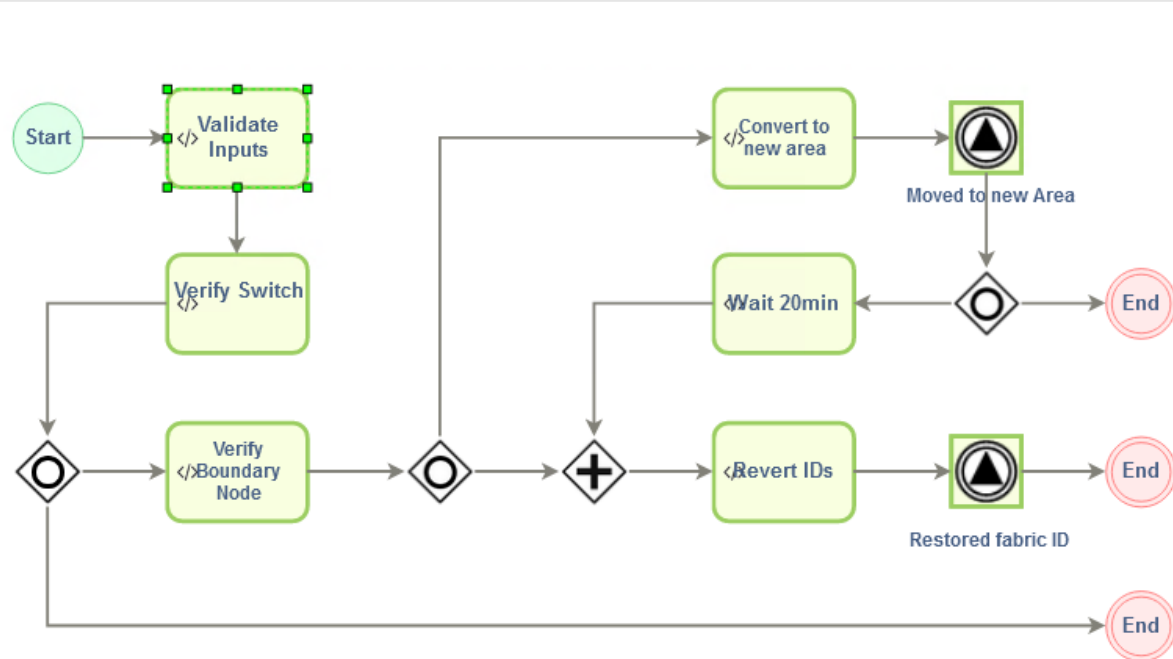
Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message	Path
✓	12/8/2022 9:18:17 ...	Fabric MultiArea Mi...	52	Workflow Designer ...	2	Istevens	12/8/2022 9:18:45 ...	Configured RADIUS and EAPoL on devices 20.0.20.31, 20.0.20.32	/Workflows/Ludovico/Fabric MultiArea Migrate

Graph View Table View



Stop Workflow Show Output Show Variables



Devices Grid

Show Output

Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	20.0.20.31		12/8/2022 9:1...	12/8/2022 9:1...	Validated ...
SUCCESS	20.0.20.32		12/8/2022 9:1...	12/8/2022 9:1...	Validated ...

Run Fabric MultiArea Migrate workflow on last node to move to new area



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Notes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
20.0.20.76

Fabric ID mask manipulation Notes:
Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

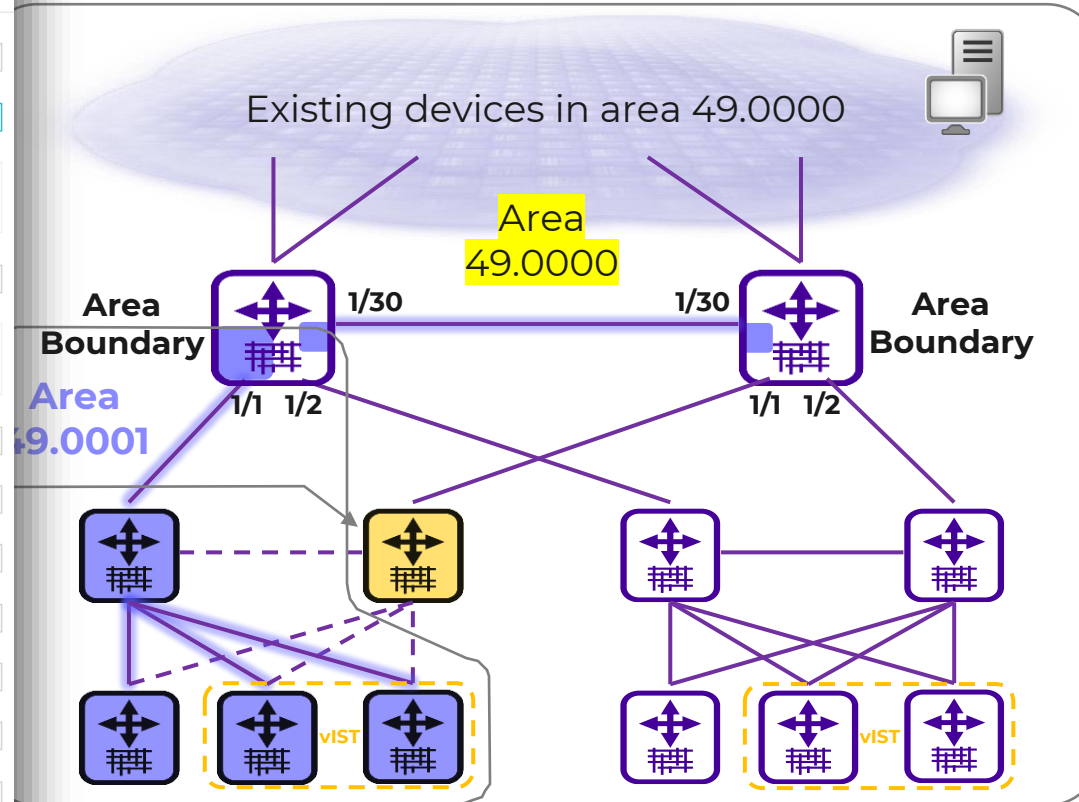
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:

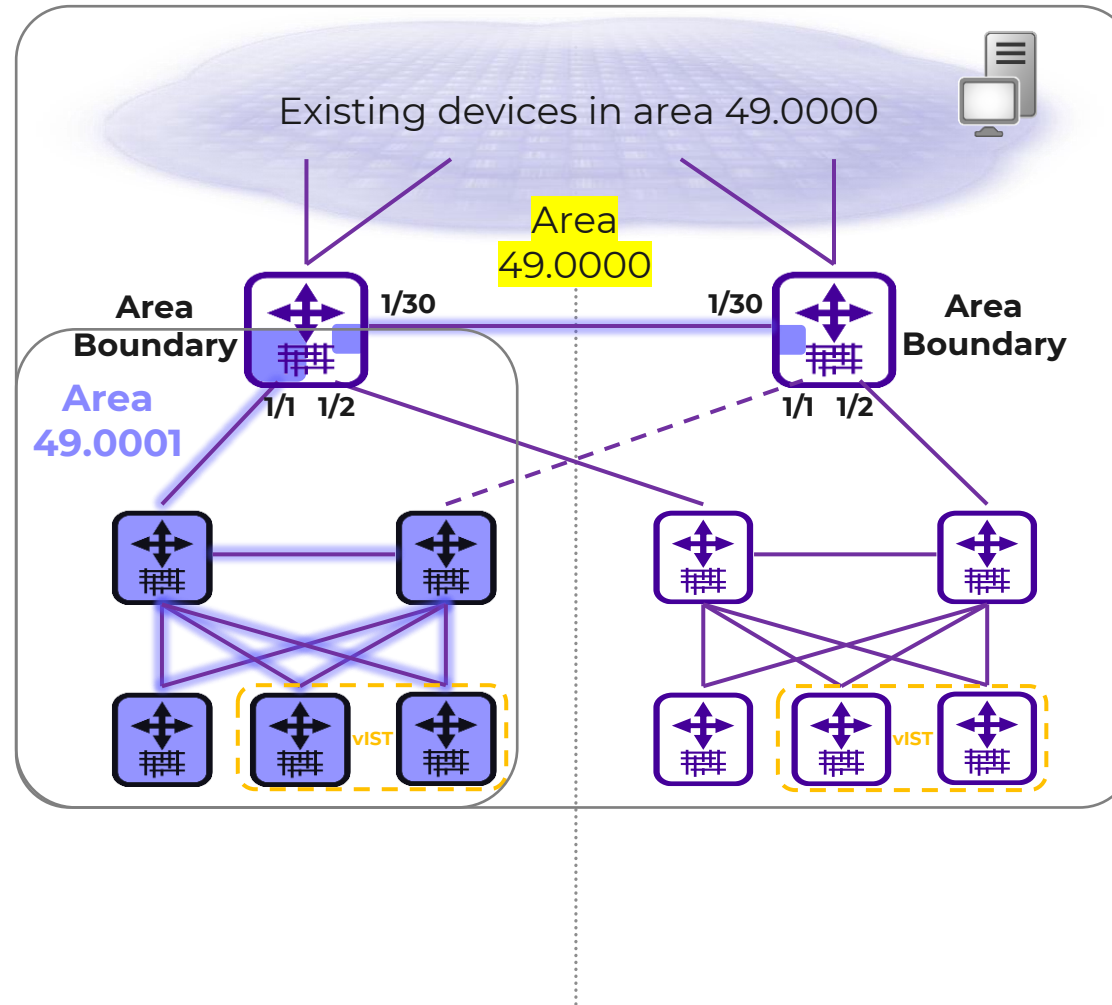


- Run the workflow against the switch
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
- Run the workflow
 - Changes are applied dynamically without a reboot

Last distribution node comes online in new access area

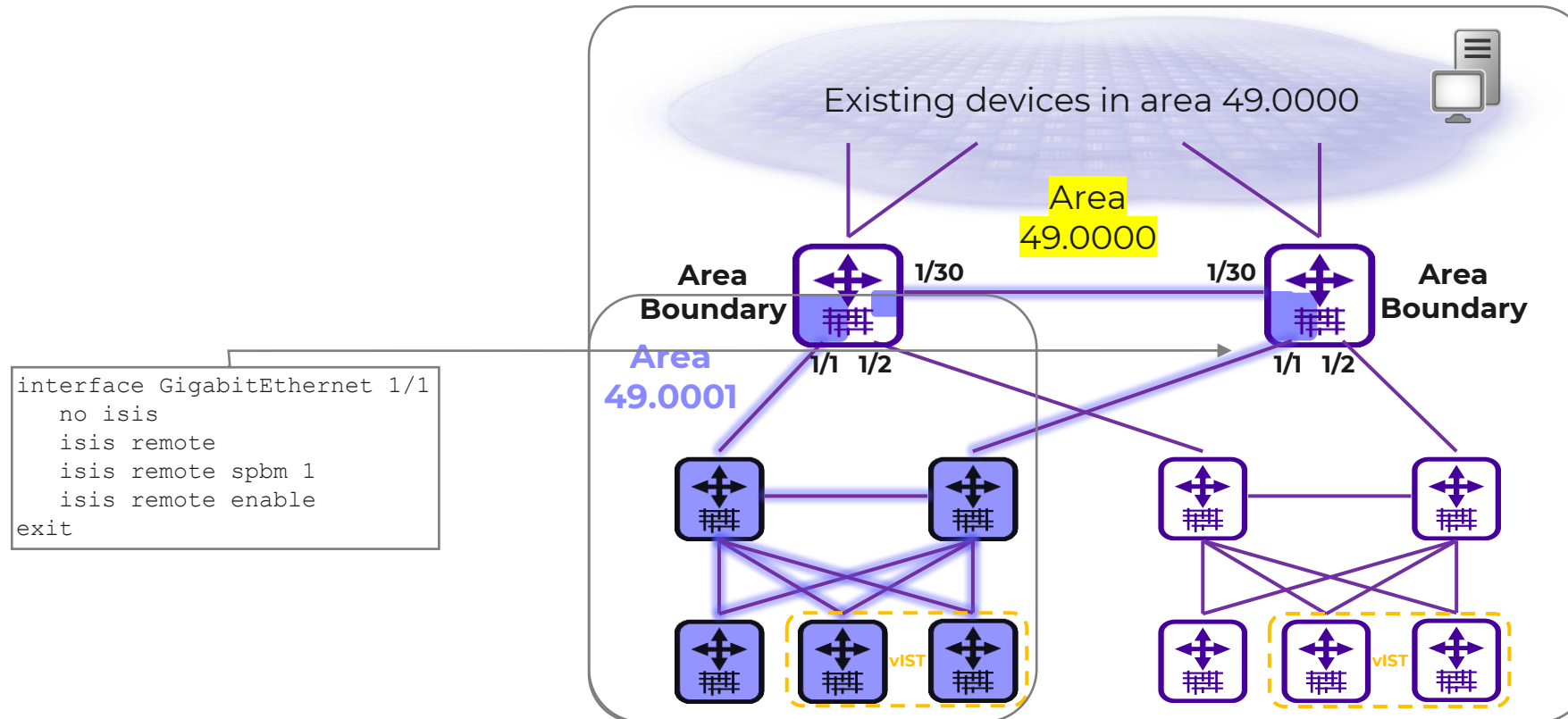


- When the last distribution node comes back online, it will have an adjacency in the new area 49.0001 to the 1st boundary node
- But its adjacency to the 2nd boundary nodes will be down, as that NNI is still using area 49.0000



- If it was desired to revert the node back to its original nickname and system-id the workflow will wait 20 minutes and then revert to the original IDs
- Should the workflow be unable to re-connect to revert the fabric IDs, it will be sufficient to manually re-run the same workflow (with same inputs) a second time and it will resume from where it left

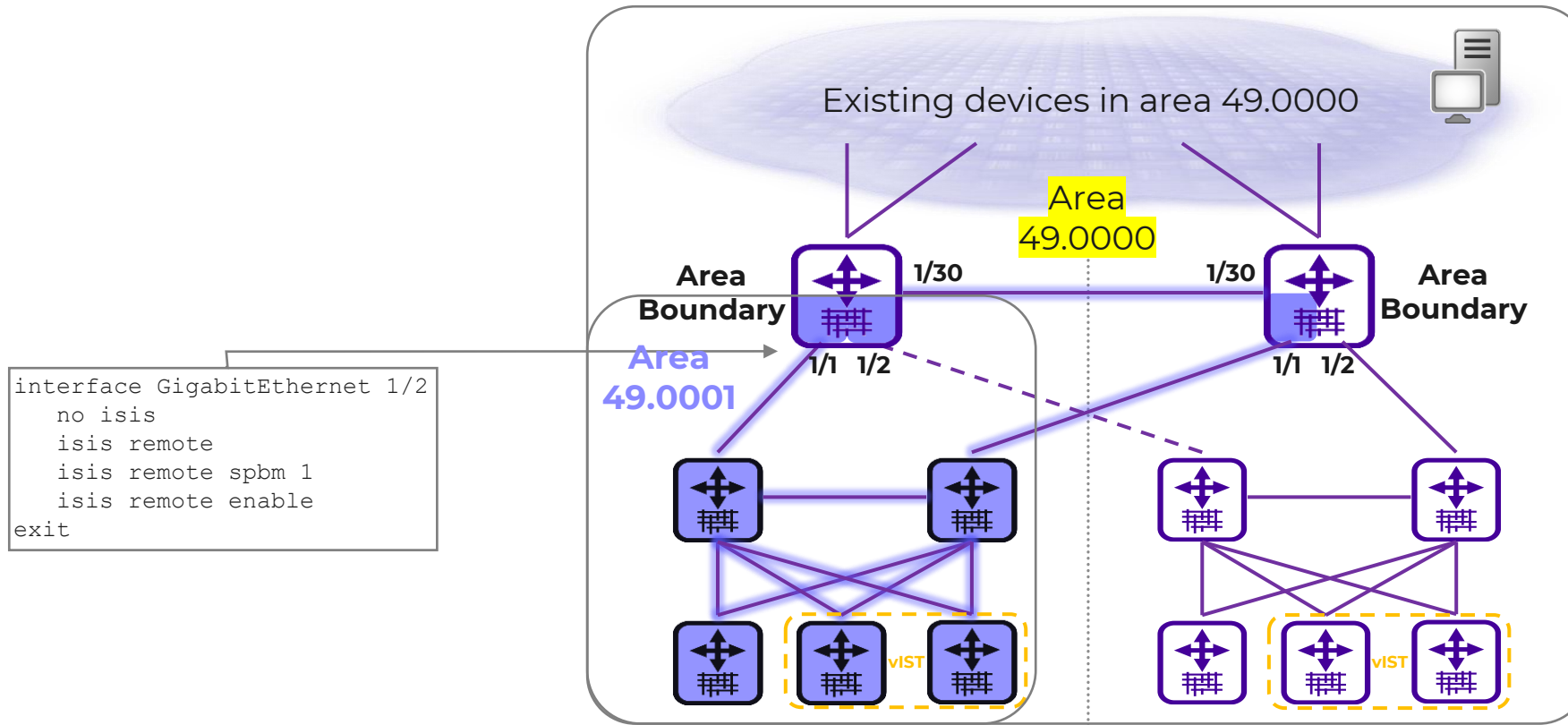
Complete boundary remote Area NNI on other boundary node



- The NNI adjacency will come up immediately, as the new area 49.0001 is already configured on the distribution node across the link

Repeating same steps

Prepare first boundary remote Area NNI



- Delete home area ISIS and re-create as remote area ISIS NNI
 - Note that it is not possible to have both ISIS Home and Remote configured on these same NNI interfaces (this can only work on direct NNI links between the two boundary nodes)
- This will cause the ISIS adjacency on the link to go down and not come back up, because the access distribution node at the other end is still configured for area 49.0000, not area 49.0001
- Access nodes can still be reached on area 49.0000 via other boundary node's NNI

Run Fabric MultiArea Migrate workflow to move to new area



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Notes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
20.0.20.76

Fabric ID mask manipulation Notes:
Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

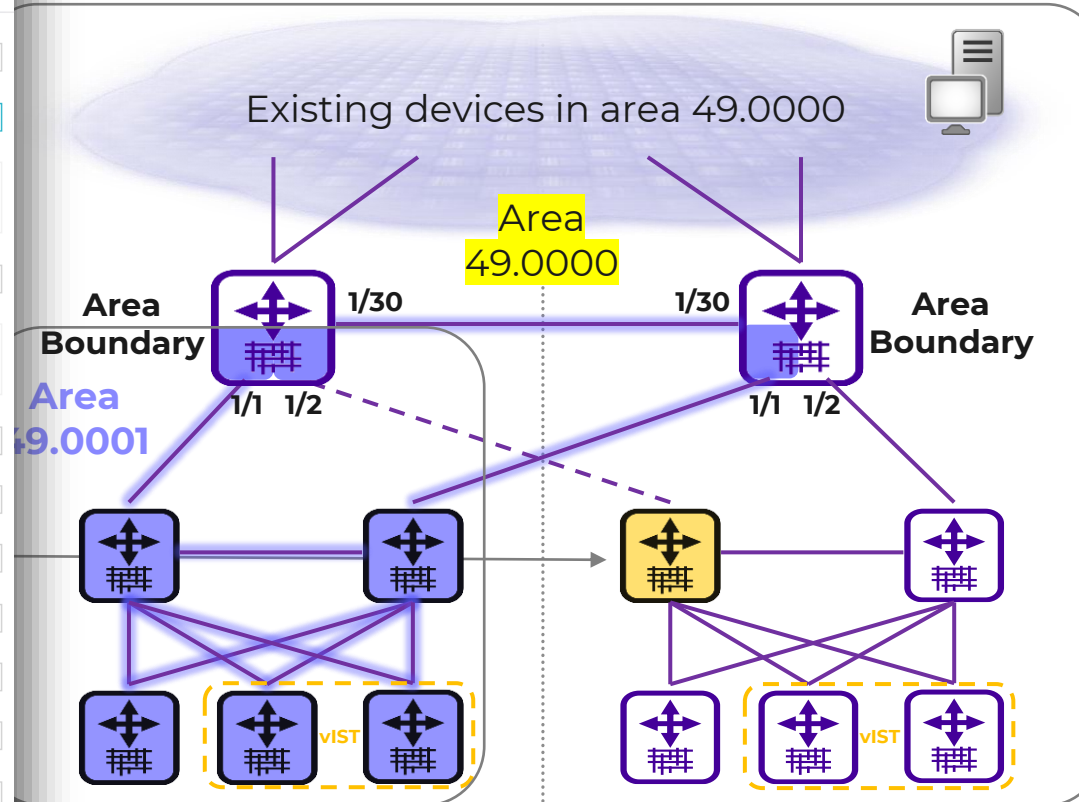
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:



- Run the workflow against the switch
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
- Run the workflow
 - Changes are applied dynamically without a reboot

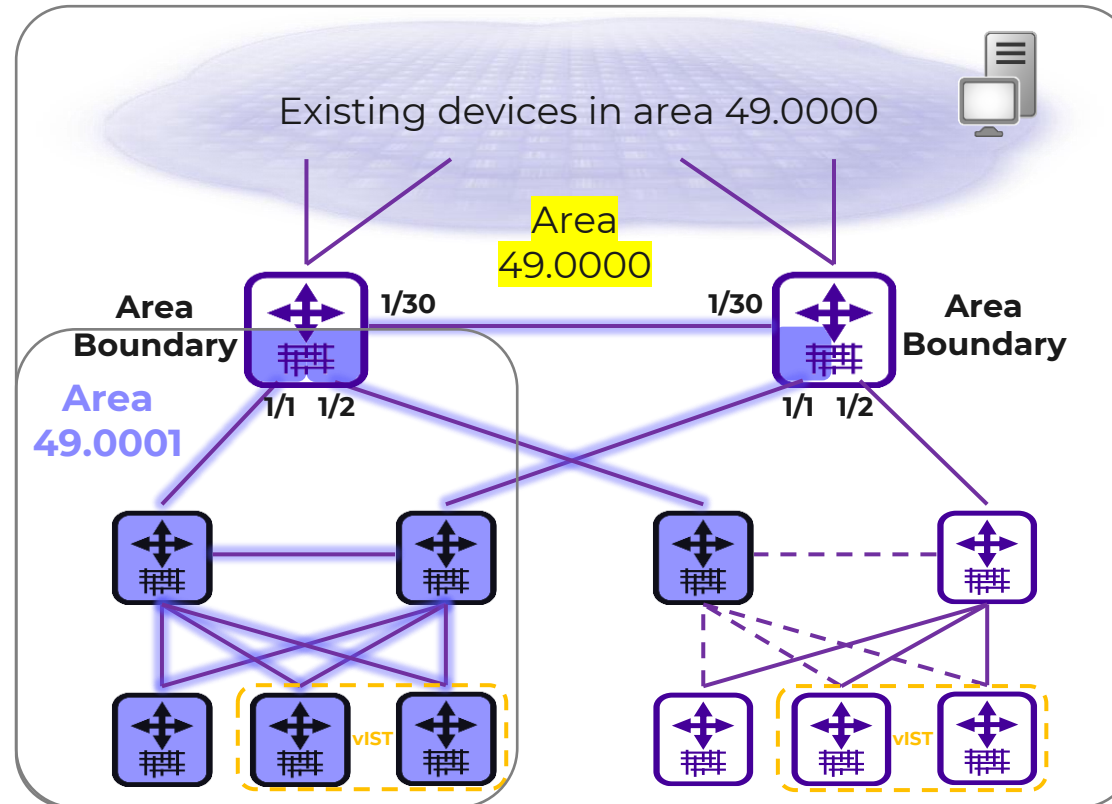
Important to realize that the node's existing nick-name and system-id are already advertised in the home area LSDB and will remain there for up to 20 minutes after the node is removed from the area 49.0000

The node must therefore be given a new nick-name and system-id to operate in the new area

1st node comes online in new access area



- When the node comes back online, it will have an adjacency in the new area 49.0001 to the boundary node
- But all other ISIS adjacency it had will be down, as its old neighbours are still in area 49.0000



Run Fabric MultiArea Migrate workflow to move to new area



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Notes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
20.0.20.76

Fabric ID mask manipulation Notes:
Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

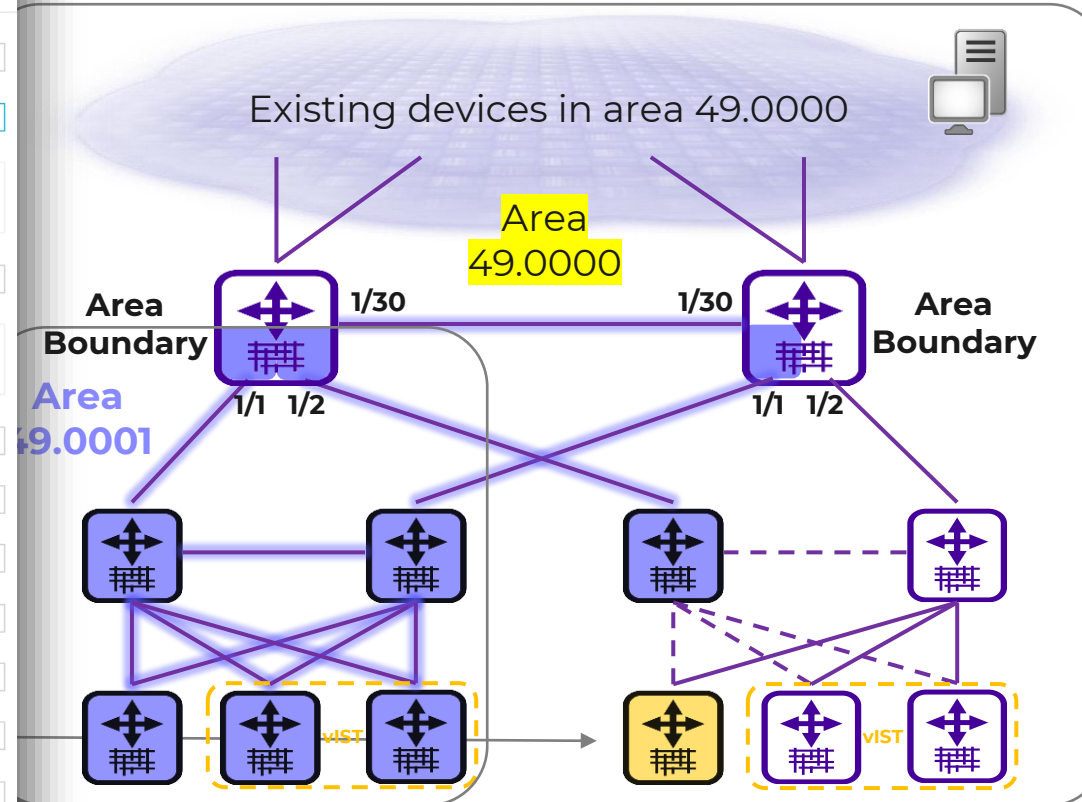
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:

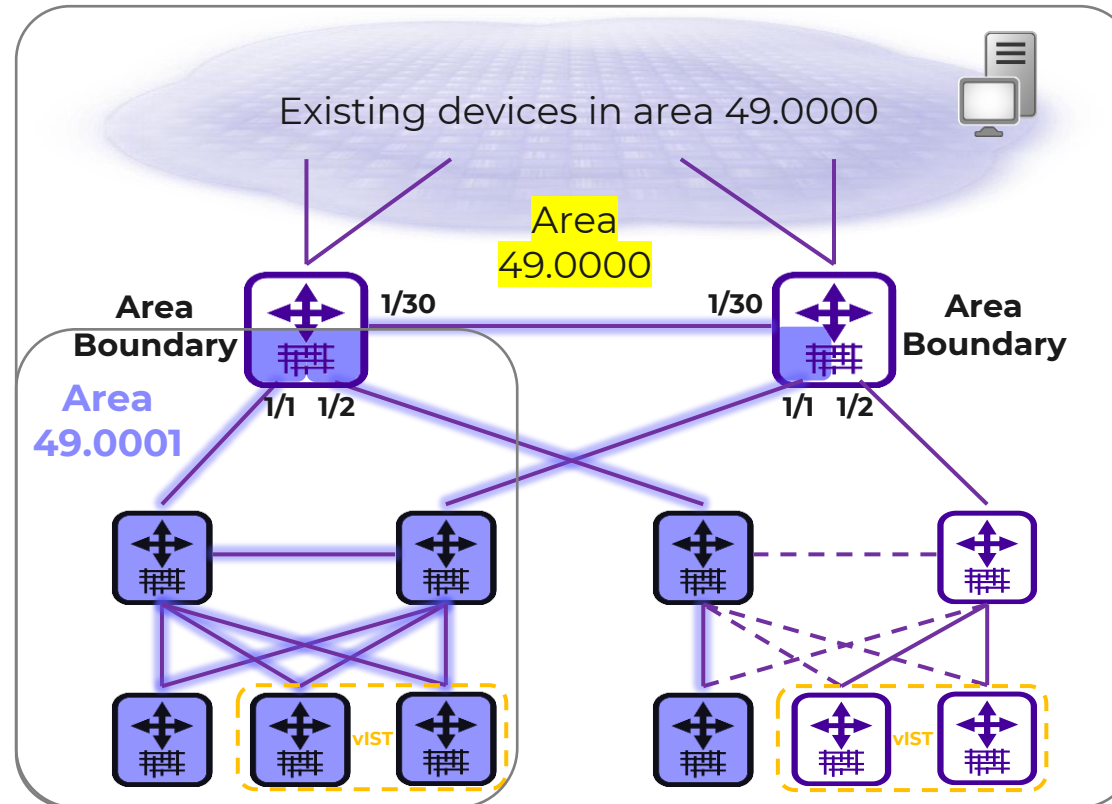


- Run the workflow against the switch
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
- Run the workflow
 - Changes are applied dynamically without a reboot

2nd node comes online in new access area



- When the node comes back online, it will have an adjacency in the new area 49.0001 going into the 1st boundary node
- But all other ISIS adjacency it had will be down, as its old neighbours are still in area 49.0000



Run Fabric MultiArea Migrate workflow to move to new area



Run Workflow - Fabric MultiArea Migrate

Workflow Inputs

Inception ISIS Area:
49.0000

Target ISIS Area:
49.0001

Mgmt IP of Area Boundary Notes:
The mgmt IP of one of the area boundary nodes connecting both ISIS areas together must be provided. This allows the workflow to verify that the fabric IDs for the node being moved will be unique in both areas, including after reverting those fabric IDs if this option is selected. No config changes will be applied on the area boundary node.

Mgmt IP of 1 Area Boundary node:
20.0.20.76

Fabric ID mask manipulation Notes:
Provide mask to identify bits to modify in the system-id and nick-name. Select the manipulation required. XOR will simply flip the masked bits (replace value field not required). Replace will replace the selected bits with provided replace value.

System-id mask manipulation:
replace

System-id mask:
0000.F000.0000

System-id mask replace value:
2

System-id in new area:
Keep modified ID

Nick-name mask manipulation:
XOR

Nick-name mask:
8.00.00

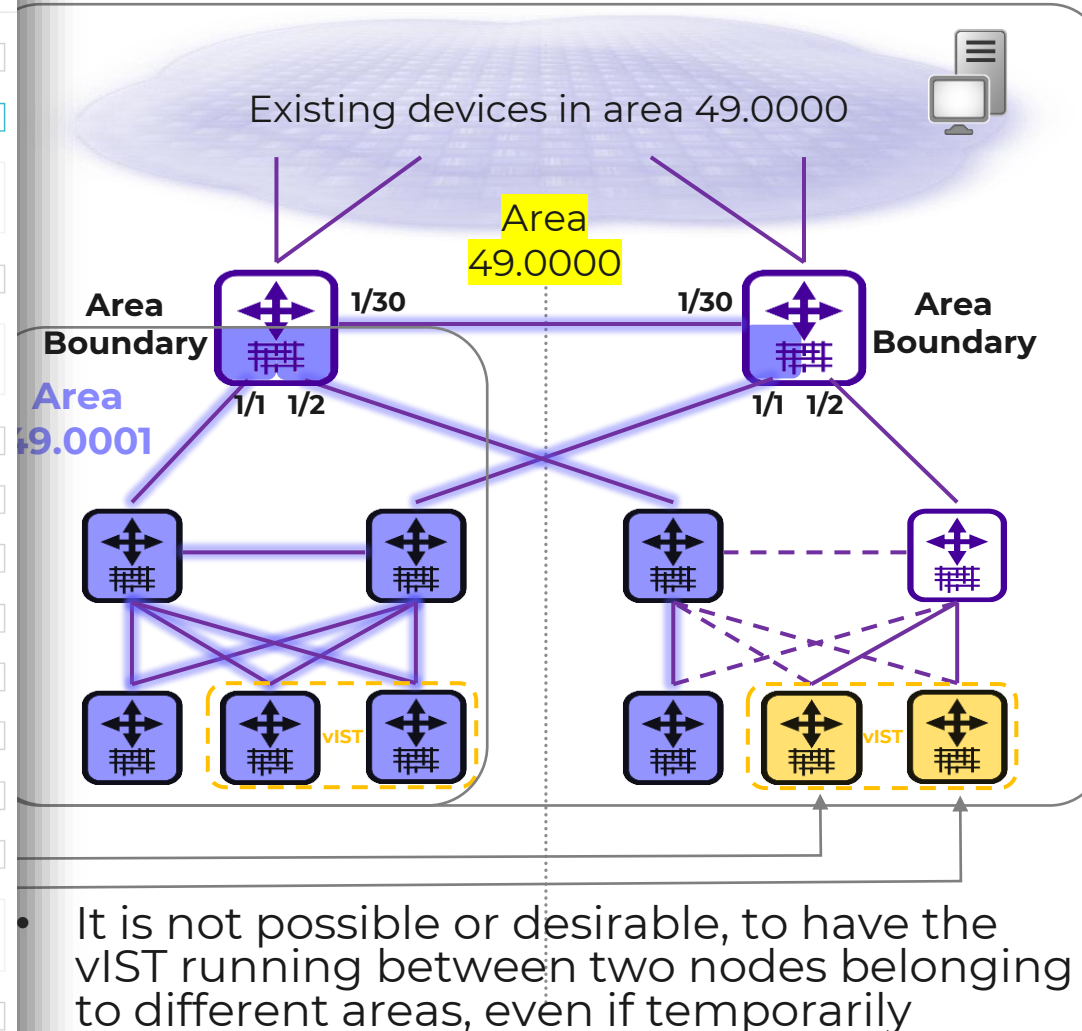
Nick-name mask replace value:

Nick-name in new area:
Revert to original ID after 20min

ISIS Hello Authentication Notes:
If it is desired to set (or change) the ISIS Hello authentication keys during the area migration, set the change pull down to "enable" and provide the new ISIS Hello auth key

Change ISIS Hello Authentication:
enable

ISIS Hello Authentication key:

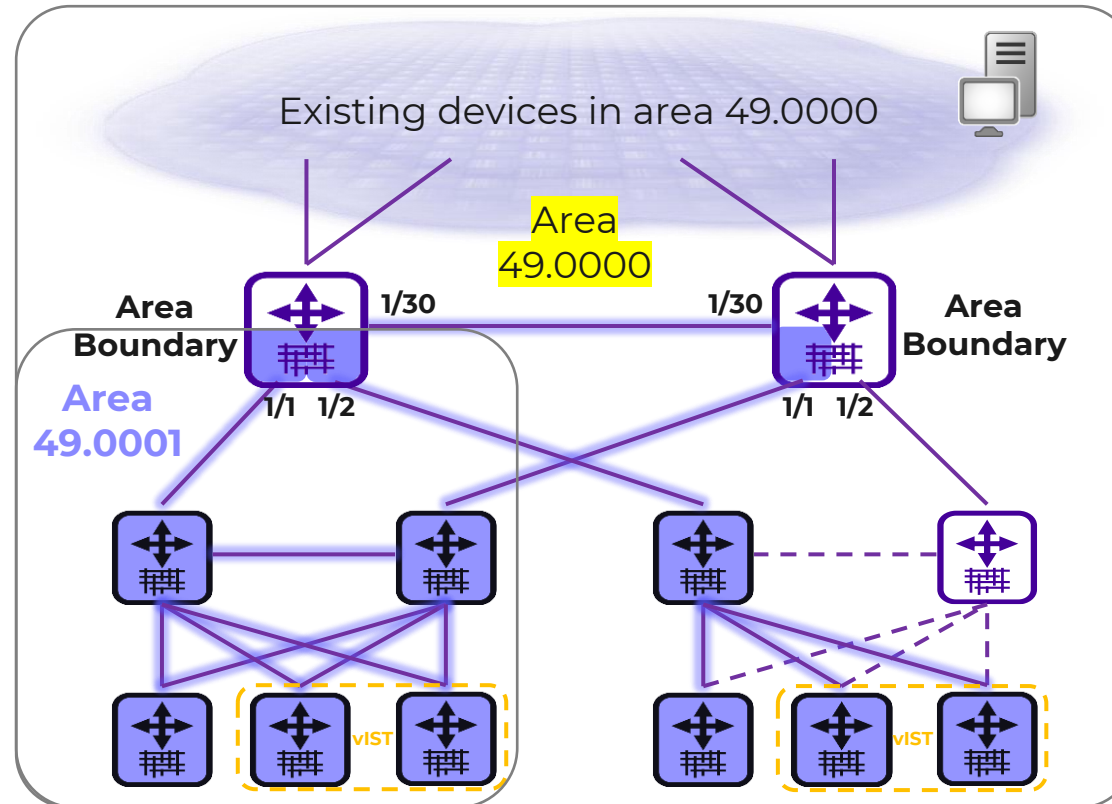


- Run the workflow against both vIST peer switches
- Provide inception and target ISIS areas
- Provide XOR or replacement manipulation to change
 - Nick-name
 - System-id
 - SMLT-Peer-Sys-id to match above
 - SMLT-Virt-BMAC
 - Manual-area
- Run the workflow
 - Changes are applied dynamically without a reboot

vIST cluster nodes come online in new access area



- When the vIST cluster comes back online, it will have adjacencies in the new area 49.0001 going into the 1st boundary node
- But all other ISIS adjacency it had will be down, as its old neighbours are still in area 49.0000



Existing devices in area 49.0000

Area Boundary

Area Boundary

Area 49.0000

1/30

1/30

1/1 1/2

1/1 1/2

VIST

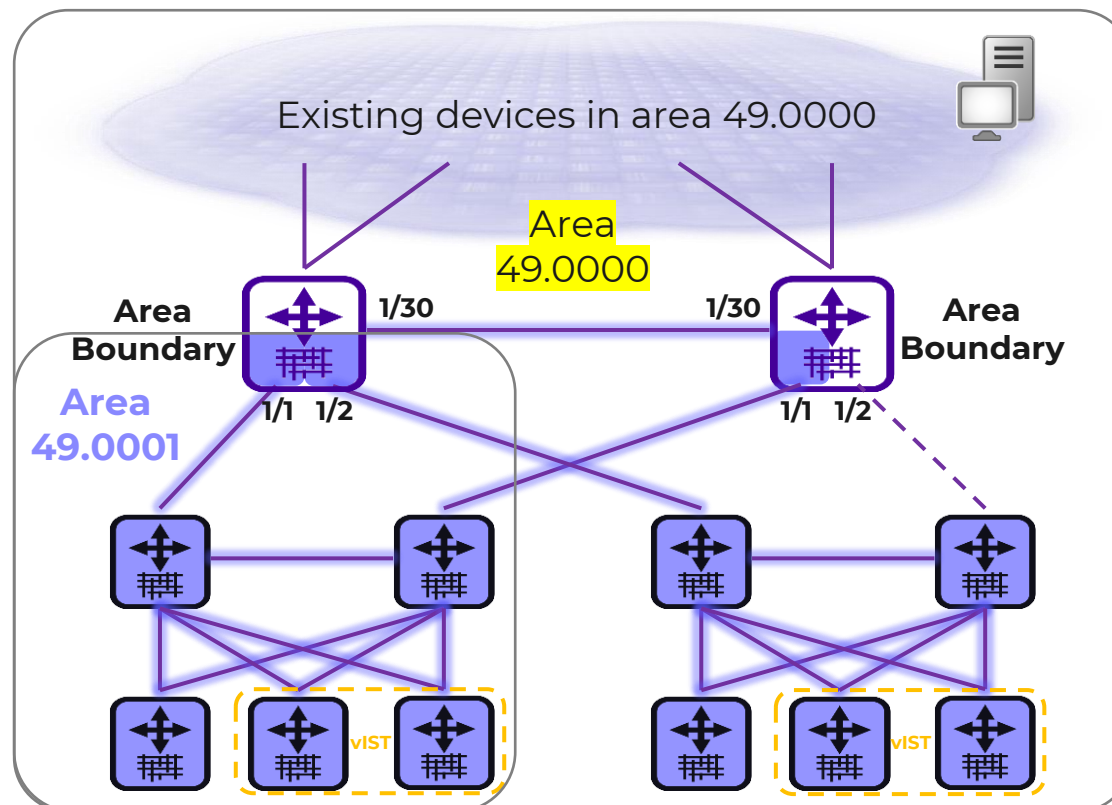
VIST

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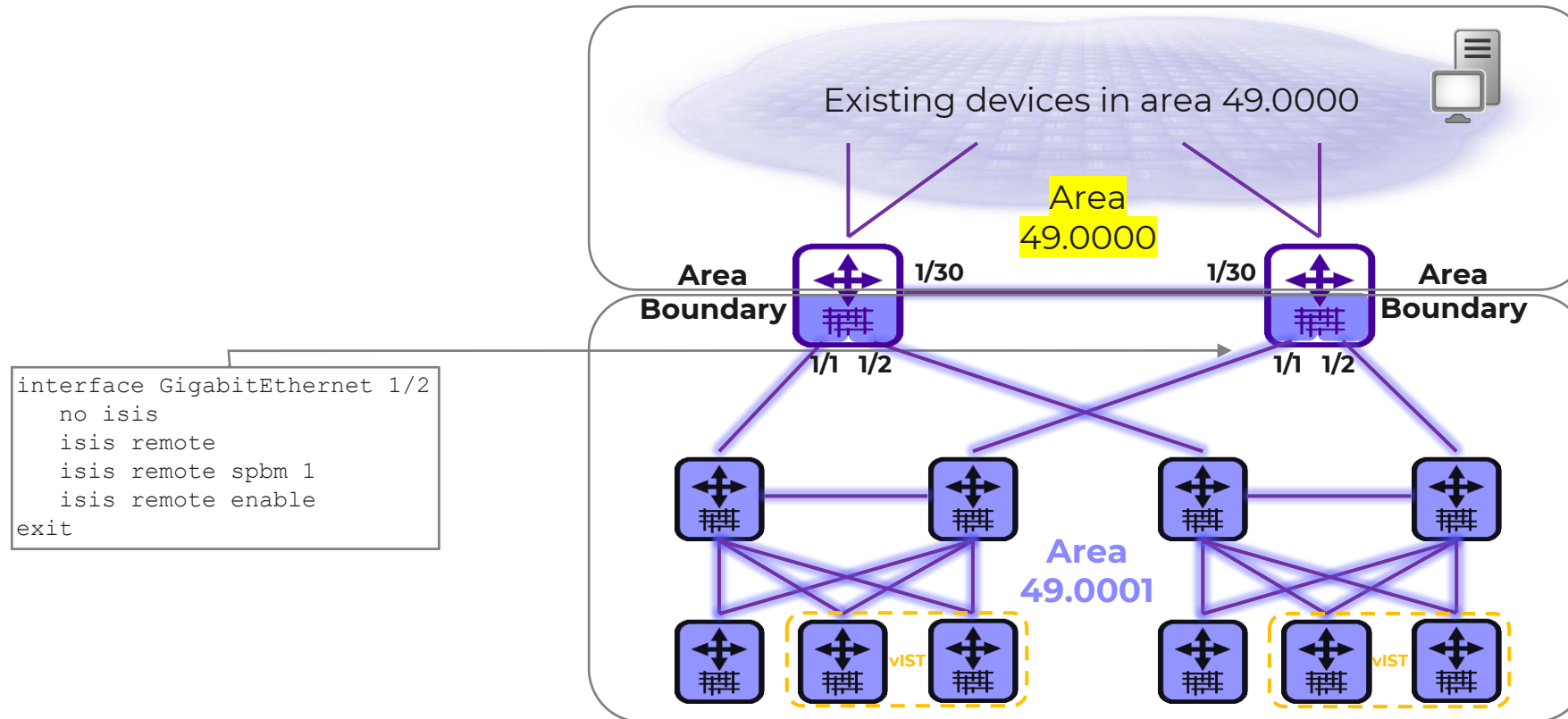
Last distribution node comes online in new access area



- When the last distribution node comes back online, it will have an adjacency in the new area 49.0001 to the 1st boundary node
- But its adjacency to the 2nd boundary nodes will be down, as that NNI is still using area 49.0000



Complete boundary remote Area NNI on other boundary node



- The NNI adjacency will come up immediately, as the new area 49.0001 is already configured on the distribution node across the link

